

# Slope-Aware LPV-MPC Path Tracking of a Dual-Steering-Wheel AGV with Time-Lag-Enhanced ModernTCN Perception

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## ABSTRACT

This paper proposes a slope-aware path-tracking method based on temporal-perception-enhanced linear parameter-varying model predictive control (LPV-MPC) to address the degradation of control accuracy and stability caused by omitting or inaccurately estimating the longitudinal road-grade angle in conventional path-tracking approaches. Firstly, to reduce reliance on potentially biased IMU-derived grade measurements, a lightweight Modern Temporal Convolutional Network (ModernTCN) is introduced to estimate the grade angle from historical onboard responses. The multi-scale time-lag offsets and the historical-window length are configured according to the vehicle response characteristics, yielding better adaptation to the operating conditions considered here than the original ModernTCN. Secondly, an uncertainty-adaptive ModernTCN-IMU interface treats the ModernTCN estimate as the primary source and uses the quality-qualified IMU estimate only as a bounded correction, thereby improving the reliability of the fused grade estimate under varying operating conditions. Furthermore, the conditioned fused-grade signal is synchronously incorporated into the LPV prediction model, measured-disturbance channel, grade-feedforward term, scheduled weights, and constraints. Compared with conventional LPV-MPC schemes that disregard grade information, this integration substantially enhances control precision and adaptation capability in graded-road scenarios. Retrospective matched simulations support improvements in lateral- and heading-error RMS; the mean control-input increment was also lower, although its paired confidence interval included zero. The proposed structure retains a lightweight temporal estimator and an exact fallback to the primary estimate.

## INDEX TERMS

Dual-steering-wheel AGV, LPV-MPC, ModernTCN, path tracking, road-grade estimation

## I. INTRODUCTION

**A**UTOMATED guided vehicles (AGVs) are widely used in smart manufacturing, automated warehousing, and industrial logistics, where they undertake material handling and flexible transport tasks [1]–[3]. Reliable AGV operation depends on several key technologies, including perception, localization, path planning, scheduling, and motion control. Among them, path tracking is the final execution layer that determines whether the planned route can be followed accurately, smoothly, and safely under actuator constraints and varying operating conditions. Therefore, robust path tracking control is a fundamental requirement for industrial AGV deployment [4]. The present study focuses on path tracking of a diagonal dual-steering-wheel (dual-drive-and-steer) AGV operating along industrial routes with longitudinal grade

changes.

Linear parameter-varying model predictive control (LPV-MPC) has been widely adopted for vehicle and AGV path tracking because it handles multivariable dynamics, reference preview, and actuator constraints within a unified optimization framework [5]–[8]. In the considered dual-steering-wheel AGV, the vehicle speed, yaw rate, and longitudinal road-grade angle form the scheduling vector. The same grade angle is also used in the measured-disturbance channel, grade-feedforward term, scheduled weights, and input bounds. Uncompensated grade variation otherwise acts as a persistent disturbance that has motivated dedicated slope-aware assistance designs [9]. Omitting or inaccurately estimating this angle may therefore cause the LPV prediction model and optimization problem to be scheduled at an oper-

ating condition inconsistent with the actual plant, resulting in prediction mismatch and degraded closed-loop tracking and control performance [10]–[12]. A straightforward approach is to estimate the grade angle using an inertial measurement unit (IMU). Yet, IMU-derived grade estimates can be biased by transient acceleration, pitch dynamics, low-cost sensor noise, and road-induced vibrations [13]–[15]. While model-based observers, such as recursive least-squares and Kalman-filter variants, can infer road grade from vehicle dynamics, wheel-speed, and driving-force data [16], [17], their accuracy depends strongly on nominal vehicle parameters, load variations, dynamic maneuvers, and the validity of wheel-ground contact assumptions [15]. These limitations motivate a data-driven temporal perception strategy. Unlike rigid physics-based observers, learning-based models can extract grade-related patterns from historical proprioceptive sequences. The coupled evolution of velocity attenuation, acceleration variation, yaw response, and drive current may therefore provide complementary information for LPV-MPC scheduling [18].

Within the realm of learning-based sequence modeling, one-dimensional convolutional neural networks (1D-CNNs) have been extensively adopted for engineering time-series analysis because of their efficiency in extracting local temporal patterns from sensor data [19]. However, when non-causal convolutional structures or offline-centered windows are directly used for online prediction, the feature representation at the current time step may be influenced by future samples, violating the causality requirement of real-time control systems. Temporal convolutional networks (TCNs) address this issue by enforcing causal convolutions, ensuring that each prediction depends only on historical and current inputs. By integrating causal structures with dilated convolutions and residual connections, TCNs effectively enlarge the receptive field and have been successfully applied to industrial sequence modeling tasks such as machinery remaining-useful-life prediction [20], [21].

Nevertheless, standard TCNs are not fully suited to industrial multivariate AGV perception. Their causal dilated convolutions enlarge the receptive field through fixed sampling intervals, but a mismatched dilation pattern may weaken short grade-sensitive transients. Meanwhile, AGV grade perception depends on coupled variables including velocity, acceleration, yaw response, and drive current, making cross-variable interaction important for reliable estimation [22]. ModernTCN revisits pure convolutional time-series modeling with large-kernel depthwise convolution for temporal dependency modeling and pointwise channel mixing for cross-variable interaction [23]. These principles provide a suitable basis for the history-only lightweight estimator developed in this work.

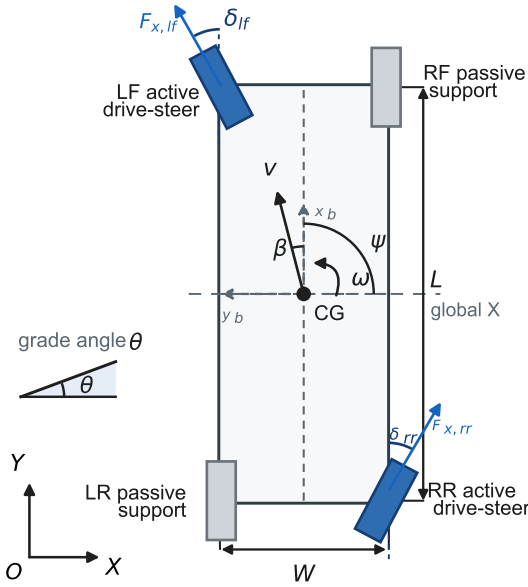
However, constructing a lightweight history-based estimator from the temporal modeling principles of ModernTCN does not by itself resolve slope-aware path tracking. Its temporal input must preserve rapid grade-sensitive vehicle responses while providing sufficient historical context to distinguish persistent road inclination from transient acceleration,

vibration, and local surface irregularities. The lag interval and historical window length therefore become distinct but coupled design choices. The problem is further complicated by the condition-dependent reliability of the learning-based and inertial estimates, as well as by the need to translate the estimated grade into control-relevant scheduling information for LPV-MPC. Learning-enhanced MPC for mobile platforms typically embeds learned components in the prediction model or controller [24], [25]; here, learning stays on the perception side. These unresolved perception-to-control issues define the scope of the contributions presented below.

The methodological contributions are threefold:

- 1) Compared with direct IMU-based grade estimation [13], [26], [27], we adapt the established temporal-modeling principles of ModernTCN [23] to infer road grade from history-only AGV responses without a direct grade input. To improve grade-estimation accuracy, the task-specific estimator combines a dynamics-informed 22-dimensional proprioceptive feature set with multi-scale temporal differences tailored to AGV dynamics and grade-signal characteristics. This representation retains operating-condition context while capturing short grade-sensitive responses, supporting more accurate grade estimation in both offline evaluation and the closed-loop simulation interface.
- 2) Compared with using our adapted ModernTCN estimator alone, whose temporal-modeling backbone is simplified from the established ModernTCN architecture [23], we develop uncertainty-adaptive fusion that uses the Qualified observer as a calibrated, innovation-conditioned, bounded complementary correction with exact fallback, improving grade-aware tracking and yielding lower mean control-input increments in the matched comparison.
- 3) Compared with LPV-MPC path tracking that omits grade scheduling [6], [28], [29], we use one conditioned fused-grade signal to synchronously update the prediction model, measured-disturbance channel, grade feedforward, scheduled weights, and constraints, enabling consistent grade-aware adaptation while retaining model-based interpretability and constraint handling.

The remainder of this paper is organized as follows. Section II establishes the model of the diagonal dual-steering-wheel AGV and formulates the slope-aware path-tracking and history-based grade-estimation problem. Section III develops the proposed framework, comprising the lightweight ModernTCN grade estimator, the dynamics-informed feature and multi-scale delta-lag representation, the uncertainty-adaptive fusion interface between ModernTCN and the Qualified observer, and the fused-grade-scheduled LPV-MPC. Section IV describes the simulation platform, the frozen experimental configuration, and the evaluation protocol. Section V reports and discusses the offline and closed-loop results. Finally, Section VI concludes the paper.



**FIGURE 1.** Configuration of the diagonal dual-steering-wheel AGV. The LF and RR wheels are active drive-steer wheels, while the RF and LR wheels are passive support wheels. The principal motion variables and road-grade angle are indicated.

## II. AGV MODELING AND HISTORY-BASED GRADE ESTIMATION

### A. AGV MODEL

The considered diagonal dual-steering-wheel AGV has active drive-steer wheels at the left-front (LF) and right-rear (RR) positions, while the right-front (RF) and left-rear (LR) wheels provide passive vertical support. The normal loads of the passive wheels are retained in the load distribution and rolling resistance, whereas their lateral-force and yaw-moment contributions are neglected. The vehicle configuration and the principal motion variables are illustrated in Fig. 1. Following the geometry-first modeling sequence used for multi-steering AGVs [30], [31], the steering geometry is established before the actuator, load, tire-force, and body-dynamics models.

The vehicle motion-state and control vectors are

$$\begin{aligned} x &= [X \ Y \ \psi \ v \ \omega \ \delta_{lf} \ \delta_{rr} \ \beta]^T, \\ u &= [F_{cmd} \ \omega_{cmd}]^T, \end{aligned} \quad (1)$$

where  $(X, Y)$  and  $\psi$  denote the position and heading in the two-dimensional road-aligned path coordinates,  $v$  is the longitudinal speed,  $\omega$  is the yaw rate,  $\delta_{lf}$  and  $\delta_{rr}$  are the active-wheel steering angles, and  $\beta$  is the vehicle sideslip angle. The longitudinal road-grade angle  $\theta$  is attached to the corresponding path station and enters the plant as an exogenous variable, with  $\theta > 0$  denoting an uphill grade along the positive direction of travel.

### 1) Kinematic Model

This study considers planar path tracking under longitudinal road grades. The reference route is represented in a two-dimensional road-aligned coordinate chart, while the grade

angle is treated as a path-dependent exogenous variable. Vertical displacement, body-pitch dynamics, road bank, and cross-slope effects are outside the present scope. Consequently, the grade angle does not directly enter the planar kinematic mapping; instead, it affects the vehicle dynamics through grade resistance, grade-dependent normal loads, and tire-force limits. The planar vehicle kinematics are therefore

$$\dot{X} = v \cos(\psi + \beta), \quad \dot{Y} = v \sin(\psi + \beta), \quad \dot{\psi} = \omega. \quad (2)$$

Here, the longitudinal grade is assumed to be locally aligned with the path tangent.

For curve motion, the instantaneous center of rotation (ICR) is assumed to lie on the body  $y$ -axis. For the smoothed speed-yaw-rate pair  $(v_{icr}, \omega_{icr})$ , its signed radius is  $R_{icr} = v_{icr} \operatorname{sgn}(\omega_{icr}) / \max(|\omega_{icr}|, \varepsilon)$ , where  $\varepsilon = 10^{-6}$ . With the center of mass midway between the axles, the active-wheel steering targets are

$$\begin{aligned} \delta_{lf}^{tgt} &= \arctan\left(\frac{L_w/2}{R_{icr} - W/2}\right), \\ \delta_{rr}^{tgt} &= -\arctan\left(\frac{L_w/2}{R_{icr} + W/2}\right), \end{aligned} \quad (3)$$

where  $L_w$  and  $W$  are the wheelbase and track width. Both targets are set to zero when  $|\omega_{cmd}| < \omega_{min}$  or  $v < v_{low}$ . A first-order actuator drives each  $\delta_i$  toward  $\delta_i^{tgt}$  with time constant  $\tau_\delta$ , subject to the rate bound  $|\dot{\delta}_i| \leq \dot{\delta}_{max}$  and angle bound  $|\delta_i| \leq \delta_{max}$ .

### 2) Dynamic Model

The motor-gearbox chain limits the drive command before it reaches the wheels. For current limit  $I_{max}$ , torque constant  $k_t$ , gear ratio  $n$ , transmission efficiency  $\eta$ , and wheel radius  $r$ , the single-wheel force limit is  $F_w^{max} = I_{max} k_t \eta n / \max(r, \varepsilon)$ . The equivalent first-order current model uses  $\tau_e = L_m / R_m$  and the effective current limit  $I_{eff} = \min(I_{max}, I_{max}^{volt})$ , where  $I_{max}^{volt} = \max[0, (V_{bus} - k_e \omega_m) / \max(R_m, \varepsilon)]$  and  $\omega_m = |v|n/r$ . The command allocation and current-to-force dynamics are

$$\begin{aligned} F_{cmd}^{eff} &= \operatorname{sat}(F_{cmd}, -2F_w^{max}, 2F_w^{max}), \\ F_{x,lf}^{cmd} &= F_{x,rr}^{cmd} = \frac{F_{cmd}^{eff}}{2}, \\ I_i^{tgt} &= \operatorname{sat}\left(\frac{F_{x,i}^{cmd} r}{k_t n \eta}, -I_{eff}, I_{eff}\right), \\ \dot{I}_i &= \frac{I_i^{tgt} - I_i}{\tau_e}, \quad F_{x,i}^{pre} = I_i \frac{k_t n \eta}{r}, \quad i \in \{lf, rr\}. \end{aligned} \quad (4)$$

The equal allocation introduces no differential drive force. The vector  $x$  in (1) contains only vehicle motion states. To distinguish it from the state advanced by the simulation, define the full nonlinear plant state as

$$\zeta = [x^T \ I_{lf} \ I_{rr} \ \zeta_f^T]^T, \quad (5)$$

where  $\zeta_f$  collects the internal states of the implemented actuator and response filters. The currents  $I_{lf}$  and  $I_{rr}$  are therefore first-order actuator states of the full nonlinear plant

rather than elements of the vehicle motion-state vector. Their responses are retained because they provide grade-sensitive measurements for the temporal estimator. The controller introduced later uses a separate four-dimensional path-error state and does not imply that the internal plant states disappear.

The road grade enters the dynamic model through the tangential and normal components of gravity. The grade, rolling, and aerodynamic resistance terms are

$$\begin{aligned} F_{slope} &= mg \sin \theta, \\ F_{roll}^{est} &= c_r mg \cos \theta, \\ F_{aero} &= \frac{1}{2} \rho_{air} C_D A v |v|, \\ F_{res}^{est} &= F_{roll}^{est} + F_{aero} + F_{slope}. \end{aligned} \quad (6)$$

where  $\rho_{air}$  is the air density and  $C_D A$  is the product of the drag coefficient and frontal area. Thus,  $mg \sin \theta$  directly changes the longitudinal force balance, whereas  $mg \cos \theta$  changes the normal-load-dependent resistance and tire-force capacity. Based on the pre-clipping drive forces, the acceleration estimate, load transfers, and front-rear base loads are

$$\begin{aligned} a_{long} &= \text{sat} \left( \frac{F_{x,lf}^{pre} + F_{x,rr}^{pre} - F_{res}^{est}}{m}, -a_{max}, a_{max} \right), \\ \Delta_{long} &= \frac{ma_{long} h_{cg}}{L_w}, \quad \Delta_{lat} = \frac{m|v\omega| h_{cg}}{\max(W, \varepsilon)}, \\ N_f^0 &= \frac{mg \cos \theta}{2} - \Delta_{long}, \quad N_r^0 = \frac{mg \cos \theta}{2} + \Delta_{long}. \end{aligned} \quad (7)$$

Here,  $N_f^0$  and  $N_r^0$  denote total front- and rear-axle loads, respectively. With the lateral transfer shared equally between the two axles, the individual wheel loads before the nonnegative numerical safeguard are

$$\begin{aligned} \tilde{N}_{lf} &= \frac{N_f^0}{2} - \frac{\text{sgn}(\omega) \Delta_{lat}}{2}, \quad \tilde{N}_{rf} = \frac{N_f^0}{2} + \frac{\text{sgn}(\omega) \Delta_{lat}}{2}, \\ \tilde{N}_{lr} &= \frac{N_r^0}{2} - \frac{\text{sgn}(\omega) \Delta_{lat}}{2}, \quad \tilde{N}_{rr} = \frac{N_r^0}{2} + \frac{\text{sgn}(\omega) \Delta_{lat}}{2}. \end{aligned} \quad (8)$$

The implementation projects  $\tilde{N}$  onto the nonnegative simplex with total load  $mg \cos \theta$  before calculating  $F_{roll} = c_r \sum_i N_i$ . Because body-pitch dynamics are excluded, the model neglects the additional steady front-rear load redistribution that a pitched body could produce on a constant grade;  $\Delta_{long}$  represents dynamic longitudinal transfer only.

The wheel loads constrain both lateral and longitudinal tire forces. For the two active wheels,

$$\begin{aligned} \alpha_{lf} &= \delta_{lf} - \left( \beta + \frac{L_w \omega}{2 \max(|v|, v_{low})} \right), \\ \alpha_{rr} &= \delta_{rr} - \left( \beta - \frac{L_w \omega}{2 \max(|v|, v_{low})} \right), \\ F_{y,i} &= \text{sat}(-C_{\alpha,i} \alpha_i, -\mu N_i, \mu N_i), \\ F_{x,i}^{allow} &= \mu N_i \sqrt{\max \left[ 1 - \left( \frac{F_{y,i}}{\max(\mu N_i, \varepsilon)} \right)^2, 0 \right]}, \\ F_{x,i} &= \text{sgn}(F_{x,i}^{pre}) \min(|F_{x,i}^{pre}|, F_{x,i}^{allow}), \quad i \in \{lf, rr\}. \end{aligned} \quad (9)$$

For linearization, the hard saturation may be replaced by a smooth approximation. Let  $R(\delta_i)$  denote the planar rotation from wheel to body coordinates and let  $r_{lf} = [L_w/2, W/2]^T$  and  $r_{rr} = [-L_w/2, -W/2]^T$  denote the active-wheel positions. The body-frame drive resultants are

$$\begin{aligned} F^{drv} &= \sum_{i \in \{lf, rr\}} R(\delta_i) \begin{bmatrix} F_{x,i} \\ 0 \end{bmatrix}, \\ M_z^{drv} &= \sum_{i \in \{lf, rr\}} \left( r_i \times R(\delta_i) \begin{bmatrix} F_{x,i} \\ 0 \end{bmatrix} \right)_z. \end{aligned} \quad (10)$$

Finally, let  $m_{eff} = m + 2(I_w + I_m n^2)/r^2$  account for the reflected wheel and motor inertia. Using the resistance terms in (6), the longitudinal, yaw, and sideslip dynamics are

$$\begin{aligned} \dot{v} &= \frac{F_x^{drv} - F_{roll} - F_{aero} - F_{slope}}{\max(m_{eff}, \varepsilon)}, \\ M_z &= \frac{L_w}{2} (F_{y,lf} - F_{y,rr}) + M_z^{drv} - C_{yaw} \omega, \\ \dot{\omega} &= \text{sat} \left( \frac{M_z}{\max(I_z, \varepsilon)}, -\dot{\omega}_{max}, \dot{\omega}_{max} \right), \\ \dot{\beta} &= \frac{F_{y,lf} + F_{y,rr} + F_y^{drv}}{m_{eff} \max(|v|, v_{low}/10)} - \omega - C_{\beta} \beta. \end{aligned} \quad (11)$$

At low speed, the sideslip equation is replaced by  $\dot{\beta} = -C_{\beta}^{low} \beta$ . The yaw-acceleration and sideslip-rate limits are  $\dot{\omega}_{max} = 5 \text{ rad/s}^2$  and  $|\dot{\beta}| \leq 10^\circ/\text{s}$ , respectively. Thus,  $\theta$  does not directly change the planar kinematic mapping in (2); instead,  $mg \sin \theta$  changes the longitudinal force balance and the  $\cos \theta$ -dependent wheel loads modify the available tire forces. These effects propagate through  $v$ ,  $\omega$ , and  $\beta$  to the path coordinates and provide the grade-sensitive onboard responses used by the temporal estimator. The parameter values that define the plant are listed in Table 1; the same values are shared by data generation, LPV linearization, and closed-loop evaluation.

## B. SLOPE-AWARE PATH-TRACKING OBJECTIVE AND PROBLEM FORMULATION

The objective of this work is constrained path tracking under longitudinal grade variations, rather than grade estimation as an end in itself; grade coupling has likewise motivated dedicated adaptive tracking designs for multi-wheel-drive platforms [32]. The tracking error is defined as  $e_k = [e_{y,k}, e_{\psi,k}, e_{v,k}, e_{\omega,k}^{\text{trk}}]^T$ , which collects the lateral, heading, longitudinal-speed, and yaw-rate errors. The superscript distinguishes  $e_{\omega,k}^{\text{trk}}$  from the kinematic yaw-consistency feature introduced in Section III. The control problem is stated in the general form

$$\begin{aligned} \min_{\{u_k\}} \quad & J_{\text{track}}(e, \Delta u) \\ \text{s.t.} \quad & \zeta_{k+1} = \mathcal{F}_T(\zeta_k, u_k, \theta_k), \\ & u_k \in \mathcal{U}, \quad \Delta u_k \in \Delta \mathcal{U}, \end{aligned} \quad (12)$$

where  $\mathcal{F}_T(\cdot)$  is the one-step transition of the AGV model, and  $\mathcal{U}$  and  $\Delta \mathcal{U}$  are the admissible input and input-increment sets.

**TABLE 1. Physical parameters of the diagonal dual-steering-wheel AGV plant.**

| Symbol                   | Description                             | Value                                   |
|--------------------------|-----------------------------------------|-----------------------------------------|
| $m$                      | Vehicle mass                            | 200 kg                                  |
| $I_z$                    | Yaw moment of inertia                   | 16.67 kg m <sup>2</sup>                 |
| $L_w$                    | Wheelbase                               | 2.0 m                                   |
| $W$                      | Track width                             | 0.8 m                                   |
| $h_{cg}$                 | Center-of-gravity height                | 0.5 m                                   |
| $r$                      | Wheel radius                            | 0.15 m                                  |
| $I_w$                    | Wheel rotational inertia                | 0.0135 kg m <sup>2</sup>                |
| $I_m$                    | Motor-rotor inertia                     | $1.77 \times 10^{-3}$ kg m <sup>2</sup> |
| $k_t$                    | Motor torque constant                   | 1.21 N m/A                              |
| $n$                      | Gear ratio                              | 10                                      |
| $\eta$                   | Transmission efficiency                 | 0.9                                     |
| $I_{max}$                | Per-motor current limit                 | 9.0 A                                   |
| $\tau_\delta$            | Steering time constant                  | 0.08 s                                  |
| $\delta_{max}$           | Steering-angle limit                    | 90°                                     |
| $\dot{\delta}_{max}$     | Steering-rate limit                     | 30°/s                                   |
| $a_{max}$                | Longitudinal-acceleration bound         | 1.0 m/s <sup>2</sup>                    |
| $C_\alpha$               | Cornering stiffness (each active wheel) | 3000 N/rad                              |
| $\mu$                    | Tire-road friction coefficient          | 0.8                                     |
| $c_r$                    | Rolling-resistance coefficient          | 0.015                                   |
| $\rho_{air}$             | Air density                             | 1.225 kg/m <sup>3</sup>                 |
| $C_D A$                  | Drag-area product                       | 0.5 m <sup>2</sup>                      |
| $C_{yaw}$                | Yaw damping coefficient                 | 250 N m s/rad                           |
| $C_\beta, C_\beta^{low}$ | Sideslip damping (normal, low speed)    | 0, 1.0 s <sup>-1</sup>                  |

The finite-horizon cost and constraints are detailed with the controller formulation in Section III-E. Within this problem, the grade angle is a control-relevant variable rather than an isolated geometric quantity. It changes the longitudinal force balance through  $mg \sin \theta_k$  in (6) and modifies the available tire forces through the  $\cos \theta_k$ -dependent wheel loads in (9). Scheduling the controller with an inaccurate grade therefore makes the prediction model inconsistent with the plant and degrades closed-loop tracking performance [10], [12].

A direct route to  $\theta_k$  is inertial sensing. However, IMU-derived grade estimates can be biased by transient longitudinal acceleration, body-pitch excitation, road-induced vibration, and low-cost sensor noise [14]. Model-based observers that fuse inertial and drivetrain information also remain sensitive to vehicle-parameter mismatch and load variation [15]. The controller is therefore assumed to have access to neither the true grade nor a reliable instantaneous inertial measurement. Instead, the current grade is inferred from a fixed window of onboard responses available at the sampling instant,

$$\mathcal{H}_k = \{s_{k-N_h+1}, s_{k-N_h+2}, \dots, s_k\}, \quad \hat{\theta}_k^{\text{hist}} = f_\phi(\mathcal{H}_k), \quad (13)$$

where  $s_k$  collects measured or recursively derived proprioceptive quantities. The true grade and any direct pitch-based grade measurement are excluded from the estimator input, and no sample later than the current instant  $k$  is used. The window length  $N_h$  balances the observability of grade-induced responses against the dilution of grade-transition information. Its numerical value and the underlying response-time calculation are reported with the experimental settings.

These considerations reduce slope-aware tracking to four coupled requirements: a lightweight history-based temporal estimator, dynamics-informed feature and multi-scale lag construction, uncertainty-adaptive fusion of the history-

based estimate and the Qualified observer estimate, and synchronized incorporation of the fused grade into all grade-dependent LPV-MPC updates. Section III develops these components in turn to support the tracking objective in (12).

### III. SLOPE-AWARE TEMPORAL PERCEPTION AND LPV-MPC FRAMEWORK

The formulation in Section II creates a perception-to-control chain rather than four independent additions. The first problem is to infer a control-relevant grade from onboard histories without using a direct grade channel. The second is to expose both the absolute operating state and the short grade-sensitive changes that an instantaneous input can obscure. The third is to reconcile the complementary but condition-dependent information supplied by the temporal estimator and the Qualified observer. The final problem is to inject one physically consistent grade signal into every grade-dependent LPV-MPC update. Accordingly, the proposed framework proceeds from a lightweight ModernTCN estimator, through dynamics-informed feature and delta-lag construction, to adaptive grade fusion and synchronized LPV-MPC scheduling. Figure 2 summarizes this pipeline and its interfaces: onboard measurement histories feed the temporal-perception branch and the Qualified observer branch in parallel, a single conditioned fused grade  $\theta_k^{\text{sch}}$  crosses the perception-to-control boundary, and every grade-dependent element of the LPV-MPC update is scheduled from this signal synchronously.

#### A. LIGHTWEIGHT MODERNTCN-BASED GRADE ESTIMATOR

To implement the history-to-grade mapping in (13), a lightweight task-specific estimator is developed from the temporal modeling principles of ModernTCN [23]. The estimator retains large-kernel depthwise temporal convolution, pointwise channel mixing, and residual learning, while removing the full patch-based and multi-stage hierarchy of the original general-purpose architecture. This reduction gives a fixed-window network that can be exported through ONNX and executed within the MATLAB/Simulink closed loop.

Let the network input be written generically as

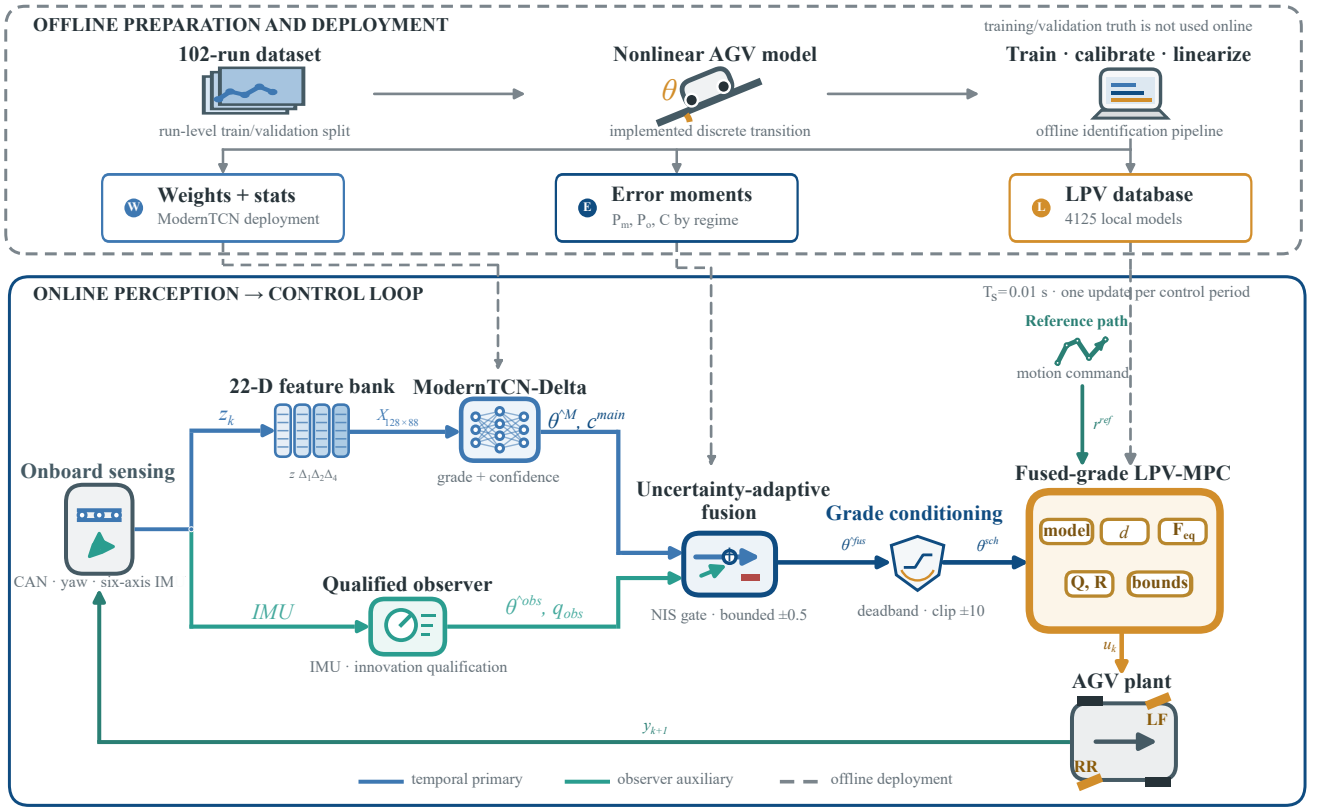
$$\mathbf{Z}_k \in \mathbb{R}^{N_h \times F}, \quad (14)$$

where its physical channels and temporal augmentation are specified in the following two subsections. A pointwise stem first projects the input to a latent channel space,

$$\mathbf{H}_k^{(0)} = \sigma[\text{BN}(\text{Conv}_{1 \times 1}(\mathbf{Z}_k^T))], \quad (15)$$

where  $\mathbf{H}_k^{(0)} \in \mathbb{R}^{C \times N_h}$ ,  $C = 64$ ,  $\text{BN}(\cdot)$  denotes batch normalization, and  $\sigma(\cdot)$  is the ReLU activation. Five stride-one residual temporal blocks are then applied. The  $\ell$ th block is

$$\begin{aligned} \mathbf{U}_k^{(\ell)} &= \text{DWConv}_K(\mathbf{H}_k^{(\ell)}), \\ \mathbf{V}_k^{(\ell)} &= \text{PW}_2^{(\ell)} \left\{ \sigma \left[ \text{PW}_1^{(\ell)} \left( \sigma \left[ \text{BN}(\mathbf{U}_k^{(\ell)}) \right] \right) \right] \right\}, \\ \mathbf{H}_k^{(\ell+1)} &= \mathbf{H}_k^{(\ell)} + \gamma_\ell \text{BN}(\mathbf{V}_k^{(\ell)}), \end{aligned} \quad (16)$$



**FIGURE 2.** Overall slope-aware perception-and-control framework. Offline preparation uses the 102-run dataset and nonlinear AGV model to deploy the ModernTCN weights and normalization statistics, fusion-error moments, and 4125-model LPV database. Online, onboard sensing drives the 22-D feature-bank/ModernTCN-Delta primary branch and the Qualified observer auxiliary branch in parallel. Their uncertainty-adaptive fusion is conditioned to form  $\theta_k^{ch}$ , which synchronously updates the LPV-MPC model, disturbance and equivalent-force terms, weights, and constraints before control of the nonlinear AGV plant. Solid blue and green paths denote the temporal-primary and observer-auxiliary signals, respectively; dashed gray paths denote offline deployment.

where  $DWConv_K$  is a depthwise temporal convolution with  $K = 31$ , the two pointwise convolutions use an expansion ratio of two, and  $\gamma_\ell$  is a learnable residual scale. The five large-kernel blocks provide a nominal receptive field of 151 samples, thereby covering the implemented historical window.

The final sequence is summarized by its last, mean, and maximum latent responses. Window-level statistics of the normalized input are added to retain low-order information about each channel:

$$\mathbf{h}_k = \left[ \mathbf{H}_{k,:;N_h}^{(5)}; \text{mean}_t \mathbf{H}_{k,:;t}^{(5)}; \max_t \mathbf{H}_{k,:;t}^{(5)}; q(\mathbf{Z}_k) \right], \quad (17)$$

$$\hat{\theta}_k^{\text{raw}} = \mathbf{w}_\theta^T \mathbf{h}_k + b_\theta,$$

where  $q(\mathbf{Z}_k)$  concatenates the current value, mean, standard deviation, maximum, and minimum of each input channel. Although same padding is used inside the fixed window,  $\mathbf{Z}_k$  terminates at the current sample and contains no measurement later than  $k$ . The deployed estimator is therefore a history-only online mapping; it is not described as a strictly causal convolutional backbone. Before fusion, the raw regression output is processed by a causal deployment interface, which applies first-order temporal smoothing, the configured physical magnitude and rate limits, and the MPC-side deadband.

The resulting control-facing estimate is denoted by  $\hat{\theta}_k^{\text{MTCN}}$  in the remainder of this section. This distinction prevents the network readout from being conflated with the stateful signal actually consumed by the fusion module.

The shared representation additionally feeds two auxiliary classification heads,

$$\begin{aligned} \hat{\mathbf{p}}_k^{\text{main}} &= \text{softmax}(g_{\text{main}}(\mathbf{h}_k)), \\ \hat{\mathbf{p}}_k^{\text{turn}} &= \text{softmax}(g_{\text{turn}}(\mathbf{h}_k)), \end{aligned} \quad (18)$$

which identify the main operating condition and steering direction. Both heads are auxiliary to grade regression and are not evaluated as independent perception contributions. The steering-direction output is retained for representation learning and diagnostics; only  $c_k^{\text{main}} = \max(\hat{\mathbf{p}}_k^{\text{main}})$  enters the deployed interface, where it calibrates the ModernTCN uncertainty used by the fusion weight. Neither predicted label directly schedules the controller or acts as a hard switching predicate. The training objective is

$$\mathcal{L} = \lambda_\theta \mathcal{L}_\theta + \mathcal{L}_{\text{main}} + \lambda_{\text{turn}} \mathcal{L}_{\text{turn}} + \mathcal{R}_\theta, \quad (19)$$

where  $\mathcal{R}_\theta$  collects the flat-road and bounded-error grade penalties. Table 2 distinguishes the implemented network from the complete ModernTCN architecture.

**TABLE 2. Structural distinction between the original ModernTCN and the lightweight task-specific estimator.**

| Component              | Original ModernTCN [23]                                                            | Lightweight task-specific estimator used in this work                                                                 |
|------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Input embedding        | Variable-independent patch embedding with an explicit variable dimension           | Joint $1 \times 1$ channel projection of the fixed historical input                                                   |
| Temporal modeling      | Large- and small-kernel depthwise convolutions with structural re-parameterization | One large-kernel depthwise convolution ( $K = 31$ ) in each residual block                                            |
| Feature mixing         | Decoupled grouped ConvFFN modules for within- and cross-variable interaction       | Two standard pointwise convolutions with an expansion ratio of two                                                    |
| Backbone hierarchy     | Patch tokens and a multi-stage representation hierarchy                            | Five repeated stride-one residual blocks with 64 latent channels                                                      |
| Readout and deployment | Task-dependent pipelines for general time-series analysis                          | Last/mean/max latent pooling plus input statistics, grade and auxiliary heads, and ONNX deployment in MATLAB/Simulink |

## B. DYNAMICS-INFORMED 22-D PROPRIOCEPTIVE FEATURE CONSTRUCTION

The temporal architecture alone does not determine whether the input contains identifiable grade information. The AGV model shows that grade first changes longitudinal resistance and available tire forces, after which its effects appear jointly in speed, acceleration, drive effort, wheel asymmetry, and yaw-steering consistency. The estimator input is therefore restricted to online onboard measurements and recursively computed response variables that represent these complementary mechanisms, consistent with longitudinal-dynamics-based estimation from standard IMU and CAN-bus signals [33]. The resulting fixed feature vector is

$$\mathbf{s}_k = [\omega_g, I_{lf}, I_{rr}, \omega_{w,lf}, \omega_{w,rr}, \delta_{lf}, \delta_{rr}, \hat{v}, \hat{v}_k, \Delta\omega_w, I_{\Sigma}, \Delta I, \Delta I_{|\cdot|}, \hat{\kappa}, a_I, \hat{v}_{LP}, a_{x,w}, I_{drv}, I_a, \Lambda, a_{HP}, \varepsilon_{\omega}^{\text{kin}}]_k^T \in \mathbb{R}^{22}. \quad (20)$$

Here,  $\omega_g$  is the measured yaw rate;  $I_{lf}$  and  $I_{rr}$  are the active-wheel motor currents;  $\omega_{w,lf}$  and  $\omega_{w,rr}$  are the corresponding wheel speeds; and  $\delta_{lf}$  and  $\delta_{rr}$  are the steering angles. The wheel-speed-derived motion responses are updated recursively as

$$\begin{aligned} \hat{v}_k &= \frac{r}{2} (\omega_{w,lf,k} + \omega_{w,rr,k}), \\ a_{x,w,k} &= \frac{\hat{v}_k - \hat{v}_{k-1}}{T_s}, \\ \hat{v}_k &= \alpha_d a_{x,w,k} + (1 - \alpha_d) \hat{v}_{k-1}, \\ \hat{v}_{LP,k} &= \alpha_{LP} \hat{v}_k + (1 - \alpha_{LP}) \hat{v}_{LP,k-1}, \\ a_{HP,k} &= a_{x,w,k} - \hat{v}_{LP,k}, \end{aligned} \quad (21)$$

where  $\alpha_d = T_s/(\tau_d + T_s)$  and  $\alpha_{LP} = T_s/(\tau_{LP} + T_s)$ . The raw, filtered, and high-pass terms retain rapid wheel-speed changes, the persistent acceleration trend, and short longitudinal transients, respectively.

At the low operating speeds considered here,  $\hat{v}$  is used only as a wheel-speed-derived motion proxy. It does not correct for steering projection or longitudinal slip and is therefore not interpreted as an exact vehicle speed during sharp steering or tire slip. The bilateral drive responses are

$$\begin{aligned} \Delta\omega_w &= |\omega_{w,lf,k} - \omega_{w,rr,k}|, \\ I_{\Sigma,k} &= |I_{lf,k}| + |I_{rr,k}|, \quad \Delta I_k = I_{lf,k} - I_{rr,k}, \\ \Delta I_{|\cdot|,k} &= |I_{lf,k}| - |I_{rr,k}|, \quad I_{drv,k} = I_{lf,k} + I_{rr,k}, \end{aligned} \quad (22)$$

which distinguish total traction demand from unequal wheel loading and turning-related allocation. The remaining consistency variables are

$$\begin{aligned} \hat{\kappa}_k &= \frac{\tan \delta_{lf,k} - \tan \delta_{rr,k}}{L_w}, \\ a_{I,k} &= \frac{\hat{v}_k}{\max(I_{\Sigma,k}, I_{\min})}, \quad I_{a,k} = \frac{I_{\Sigma,k}}{\max(|\hat{v}_{LP,k}|, a_{\min})}, \\ \Lambda_k &= \frac{I_{drv,k} - \mu_{I,tr}}{\max(\sigma_{I,tr}, \epsilon_N)} - \frac{\hat{v}_{LP,k} - \mu_{a,tr}}{\max(\sigma_{a,tr}, \epsilon_N)}, \\ \varepsilon_{\omega,k}^{\text{kin}} &= \omega_{g,k} - \hat{v}_k \hat{\kappa}_k. \end{aligned} \quad (23)$$

The protected denominators  $I_{\min} = 0.1$  and  $a_{\min} = 0.05$  prevent singular behavior at low current or acceleration. The statistics  $(\mu_{I,tr}, \sigma_{I,tr})$  and  $(\mu_{a,tr}, \sigma_{a,tr})$  are fitted on the training split, and  $\epsilon_N > 0$  protects the standardization. Thus,  $\Lambda_k$  combines two separately standardized quantities and is dimensionless; it is interpreted as an empirical drive-response co-variation feature rather than as a physical force. The denominator  $L_w$  in  $\hat{\kappa}_k$  follows the front-rear steering geometry in (3); the quantity remains a low-speed curvature proxy because slip and passive-wheel kinematics are omitted. Finally,  $\varepsilon_{\omega,k}^{\text{kin}}$  expresses yaw inconsistency and is distinct from the MPC tracking error  $e_{\omega,k}^{\text{trk}}$ . The complete 22-channel input contract assembled from these mechanisms is listed in Table 3.

The feature set excludes global pose, unmeasured sideslip and wheel normal loads, the true road grade, and direct pitch- or grade-measurement channels. The yaw-rate gyroscope remains because it measures rotational vehicle response rather than grade. Thus, the network must infer grade from the coupled response pattern of the AGV instead of learning an identity mapping from a direct grade measurement.

## C. MULTI-SCALE DELTA-LAG REPRESENTATION AND HISTORICAL WINDOW ASSEMBLY

Even with the 22-D feature set, a single sample is ambiguous because similar current and acceleration levels can be produced by grade resistance, a traction or braking command, or a steering transition. Sequence models over onboard response histories have proven effective for related road-condition identification tasks [18]. Temporal differences are therefore constructed after train-set normalization,

$$\mathbf{z}_k = \mathcal{N}(\mathbf{s}_k) = \frac{\mathbf{s}_k - \boldsymbol{\mu}_{tr}}{\boldsymbol{\sigma}_{tr}}, \quad (24)$$

**TABLE 3. Dynamics-informed 22-dimensional proprioceptive feature set.**

| Feature group                          | Dim. | Implemented channels                                                                       | Primary information represented                                                 |
|----------------------------------------|------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Raw onboard sensing                    | 7    | gyro_z, I_lf, I_rr, omega_wheel_lf, omega_wheel_rr, delta_lf, delta_rr                     | Measured yaw, drive effort, wheel motion, and steering state                    |
| Kinematic and multi-band response      | 5    | v_hat, dv_hat_dt, dv_hat_dt_lp, accel_x_wheel, a_hp                                        | Current speed, persistent acceleration trend, and rapid longitudinal transients |
| Drive asymmetry and allocation         | 5    | ws_imbalance, I_sum, I_diff_signed, I_diff_abs, I_drive_signed                             | Total traction demand and unequal active-wheel response                         |
| Load and kinematic consistency proxies | 5    | kappa_proxy, accel_per_current, current_per_accel, drive_load_proxy, yaw_consistency_error | Steering curvature, drive-acceleration coupling, and yaw inconsistency          |

where  $\mu_{tr}$  and  $\sigma_{tr}$  are fitted only on the training split. For a lag  $\ell$ , the normalized delta channel is

$$\Delta_{\ell} \mathbf{z}_k = \text{clip}(\mathbf{z}_k - \mathbf{z}_{k-\ell}, -c_{\Delta}, c_{\Delta}), \quad \ell \in \{1, 2, 4\}, \quad (25)$$

with  $c_{\Delta} = 5$ . At  $T_s = 0.01$  s, the three offsets represent 10, 20, and 40 ms response changes. The augmented vector is

$$\tilde{\mathbf{z}}_k = [\mathbf{z}_k^T \quad (\Delta_1 \mathbf{z}_k)^T \quad (\Delta_2 \mathbf{z}_k)^T \quad (\Delta_4 \mathbf{z}_k)^T]^T \in \mathbb{R}^{88}. \quad (26)$$

The augmented representation in (26) is referred to as the delta bank. The current normalized state retains the absolute operating condition, whereas the three difference blocks expose increasingly wider local changes. Calculating the differences after normalization prevents the wheel-speed and current scales from dominating the augmentation, and clipping limits isolated transients. The lag set was configured through a staged representation study: single offsets from  $\{1, 2, 4, 8, 16, 32, 64\}$  were first screened, after which the short offsets  $\{1, 2, 4\}$  were carried forward. Raw input, a single-difference branch, multi-lag state stacking, the three-branch delta bank, and mixed lag-delta inputs were then compared under the same estimator and synchronized deployment interface. The deployed delta bank is evaluated through the matched-seed offline comparison in RQ2 and the closed-loop grade-estimation comparison in RQ3.

The 88-dimensional vectors are placed in chronological order,

$$\mathbf{Z}_k = \begin{bmatrix} \tilde{\mathbf{z}}_{k-N_h+1}^T \\ \tilde{\mathbf{z}}_{k-N_h+2}^T \\ \vdots \\ \tilde{\mathbf{z}}_k^T \end{bmatrix} \in \mathbb{R}^{N_h \times 88}, \quad N_h = 128. \quad (27)$$

At the 0.01-s sampling interval, the implemented window covers 1.28 s. Its length is fixed from the characteristic response time over which grade-induced resistance produces an observable change in the onboard responses at the representative speed, grade, and effective inertia of the present AGV; the corresponding parameter calculation is given in the experimental setup. For unavailable lagged samples at the beginning of a window, the first normalized sample is used for edge padding, which makes the corresponding differences zero. The same normalization, padding, endpoint labeling, and channel order are used during dataset construction, validation, ONNX export, and online inference.

#### D. UNCERTAINTY-ADAPTIVE GRADE FUSION WITH THE QUALIFIED OBSERVER

ModernTCN-delta and the Qualified observer provide complementary information for road-grade perception. ModernTCN-delta remains the primary source because it estimates grade from history-based vehicle responses, whereas the Qualified observer contributes an auxiliary inertial observation. The observer follows the low-dynamic MEMS-IMU estimation structure of [34] and adopts innovation-based adaptive-noise modulation in the spirit of [17]. In the present implementation, the observer receives the bias-compensated six-axis packet  $[\tilde{f}_x, \tilde{f}_y, \tilde{f}_z, \tilde{g}_x, \tilde{g}_y, \tilde{g}_z]^T$  at  $T_s = 0.01$  s and provides a grade-related output for the fusion interface.

The observer propagates a pitch-related state and gyroscope-bias state from the angular-rate measurements and updates them with the accelerometer-derived gravity observation. Its innovation is

$$\mathbf{r}_k^a = \mathbf{z}_k^a - h_{\text{obs}}(\mathbf{x}_k^-), \quad \mathbf{x}_k^- = f_{\text{obs}}(\mathbf{x}_{k-1}, \tilde{\omega}_k), \quad (28)$$

where  $\mathbf{z}_k^a = [\tilde{f}_x, \tilde{f}_y, \tilde{f}_z]^T$ , and  $f_{\text{obs}}(\cdot)$  and  $h_{\text{obs}}(\cdot)$  denote the bias-compensated propagation and gravity-observation maps. The observation covariance is modulated by normalized acceleration-norm, vibration, and innovation scores,

$$\begin{aligned} R_k &= R_0 \text{clip}(m_k, 1, m_{\max}), \\ m_k &= \mathcal{F}(w_n s_k^n, w_v s_k^v, w_{\nu} s_k^{\nu}), \end{aligned} \quad (29)$$

where  $(w_n, w_v, w_{\nu}) = (0.50, 0.15, 0.35)$ , the vibration persistence factor is 0.95, and  $m_{\max} = 60$ . The deployed normalization scales are 0.2g, 0.35 rad/s, and  $4^\circ$  for the acceleration norm, angular rate, and innovation, respectively. The initial pitch standard deviation is  $2^\circ$ , the initial gyroscope-bias standard deviation is 0.01 rad/s, the gyroscope-noise standard deviation is 0.002 rad/s, the bias random-walk standard deviation is  $5 \times 10^{-5}$  rad/s, and the accelerometer-derived pitch standard deviation is  $1.25^\circ$ . The resulting observer state is converted to the grade interface

$$\hat{\theta}_k^{\text{obs}} = \mathcal{G}_{\theta}(\mathbf{x}_k^{\text{obs}}), \quad (30)$$

where  $\mathcal{G}_{\theta}(\cdot)$  denotes the road-aligned grade conversion.

Fusion uncertainty is calibrated offline from the training and validation runs. Ground-truth grade is used only for this calibration and is absent from the deployed interface. The calibration stores regime-dependent total error moments for ModernTCN-delta and the Qualified observer, including their cross-error moment. At runtime, the ModernTCN confidence

and the observer quality score adjust these moments before the innovation test. With

$$\nu_k = \hat{\theta}_k^{\text{obs}} - \hat{\theta}_k^{\text{MTCN}},$$

the innovation variance, NIS, and raw gain are

$$S_k = \max(P_{M,k} + P_{O,k} - 2C_k, S_{\min}), \quad (31)$$

$$\text{NIS}_k = \nu_k^2 / S_k, \quad K_k^{\text{raw}} = (P_{M,k} - C_k) / S_k.$$

The effective gain is

$$K_k^{\text{eff}} = \eta_k \text{clip}(K_k^{\text{raw}}, 0, 1),$$

$$\eta_k = \begin{cases} 1, & \text{NIS}_k \leq 9, \\ \sqrt{9/\text{NIS}_k}, & 9 < \text{NIS}_k \leq 25, \\ 0, & \text{NIS}_k > 25. \end{cases} \quad (32)$$

The selected configuration uses  $K_{\max} = 1.0$ , a helpfulness threshold of 0.6, slope-entry and slope-exit thresholds of  $0.5^\circ$  and  $0.3^\circ$ , respectively, a minimum innovation of  $0.25^\circ$ , and an innovation rejection gate of  $2^\circ$ . Equation (32) defines the gain within an eligible correction state; otherwise, the exact identity fallback is used.

The target correction and its causal update are

$$\delta\theta_k^* = \text{clip}[K_k^{\text{eff}} \nu_k, -0.5^\circ, 0.5^\circ],$$

$$\delta\theta_k = \delta\theta_{k-1} + \text{clip}[\delta\theta_k^* - \delta\theta_{k-1}, -5^\circ \text{s}^{-1} T_s, 5^\circ \text{s}^{-1} T_s],$$

$$\hat{\theta}_k^{\text{fus}} = \hat{\theta}_k^{\text{MTCN}} + \delta\theta_k. \quad (33)$$

If the temporal estimate is not ready, the Qualified observer is unavailable, the helpfulness or slope-hysteresis condition is not satisfied, the innovation is below the minimum activation level or exceeds  $2^\circ$ , or  $\text{NIS}_k > 25$ , the exact identity fallback is applied:  $\hat{\theta}_k^{\text{fus}} = \hat{\theta}_k^{\text{MTCN}}$  and  $\delta\theta_k = 0$ . Before entering the controller, the fused grade is processed by the  $1.3^\circ$  deadband, clipped to  $[-10^\circ, 10^\circ]$ , and passed through the causal scheduling filter. The resulting  $\theta_k^{\text{sch}}$  is the single grade signal supplied to the LPV prediction model, measured-disturbance channel, grade feedforward, scheduled weights, and constraints. It is defined by

$$\theta_k^{\text{sch}} = \mathcal{F}_\rho(\theta_k^{\text{guard}}), \quad (34)$$

where  $\mathcal{F}_\rho(\cdot)$  denotes the unchanged causal scheduling filter.

Algorithm 1 summarizes the online execution order.

### E. FUSED-GRADE-SCHEDULED DISCRETE LPV-MPC

The fusion module becomes control-relevant only when its output modifies the prediction model and optimization problem consistently. The nonlinear plant is therefore converted to a discrete LPV prediction model using the same actuator update and numerical transition as the closed-loop simulation, after which the conditioned grade in (34) synchronously schedules the model, measured-disturbance path, feedforward term, weights, and bounds, following the established LPV-MPC practice of offline model construction with online scheduled updates [35].

**Algorithm 1** Online grade fusion assisted by the Qualified observer

**Require:** ModernTCN-delta estimate and confidence; current six-axis inertial packet

**Ensure:** scheduled grade  $\theta_k^{\text{sch}}$

- 1: Propagate and update the Qualified observer using (28)–(29)
- 2: Compute  $\hat{\theta}_k^{\text{obs}}$ ,  $\nu_k$ ,  $\text{NIS}_k$ , and  $K_k^{\text{eff}}$
- 3: **if** any readiness, helpfulness, slope, or innovation-consistency gate fails **then**
- 4:   Apply exact identity fallback
- 5: **else**
- 6:   Apply the bounded, rate-limited correction in (33)
- 7: **end if**
- 8: Apply the amplitude guard and causal scheduling filter
- 9: **return**  $\theta_k^{\text{sch}}$

### 1) Discrete Path-Error Model

The controller state, input, and measured grade disturbance are

$$x_k^e = [e_{y,k} \quad e_{\psi,k} \quad e_{v,k} \quad e_{\omega,k}^{\text{trk}}]^T, \quad (35)$$

$$u_k = [F_{\text{cmd},k} \quad \omega_{\text{cmd},k}]^T, \quad d_k = \theta_k^{\text{sch}}.$$

Let  $\Phi_{T_s}$  denote the implemented one-step transition of the full state  $\zeta$  in (5), including steering-rate and steering-angle saturation, midpoint steering values, fourth-order Runge–Kutta integration, angle wrapping, non-negative-speed protection, and sideslip limiting. The lifting map  $\mathcal{L}_e(x^e; r^{\text{ref}}, \bar{\zeta}_{\text{int}})$  reconstructs the absolute motion state from the path-error coordinates and the local reference pose, speed, and yaw rate, and inserts the steering, sideslip, current, and filter states  $\bar{\zeta}_{\text{int}}$  associated with the local operating point. Conversely,  $q(\zeta; r^{\text{ref}})$  maps the propagated absolute state back to the four path errors. The implemented reduced transition is therefore

$$g(x^e, u, \theta) = q(\Phi_{T_s}[\mathcal{L}_e(x^e; r^{\text{ref}}, \bar{\zeta}_{\text{int}}), u, \theta], r^{\text{ref}}, +). \quad (36)$$

During nonlinear simulation, all components of  $\zeta$  are advanced. During offline database construction, the reduced four-state model fixes the unrepresented internal states at their local operating values before finite differentiation. This makes the reduction and its approximation explicit. Define the operating-point deviations

$$\tilde{x}_k^e = x_k^e - \bar{x}^e, \quad \tilde{u}_k = u_k - \bar{u}, \quad \tilde{\theta}_k = \theta_k - \bar{\theta}, \quad (37)$$

whereas the control increment is reserved for the adjacent-sample difference  $\Delta u_k = u_k - u_{k-1}$ . The local deviation model is

$$\tilde{x}_{k+1}^e = A_d(\mathbf{q}_k) \tilde{x}_k^e + B_d(\mathbf{q}_k) \tilde{u}_k + E_d(\mathbf{q}_k) \tilde{\theta}_k,$$

$$\tilde{y}_k = C_d \tilde{x}_k^e + D_d \tilde{u}_k, \quad \mathbf{q}_k = [v_k \quad \omega_k \quad \theta_k^{\text{sch}}]^T, \quad (38)$$

where  $A_d \in \mathbb{R}^{4 \times 4}$ ,  $B_d \in \mathbb{R}^{4 \times 2}$ ,  $E_d \in \mathbb{R}^{4 \times 1}$ ,  $C_d = I_4$ , and  $D_d = 0$ . The matrices are evaluated directly from the one-step transition,

$$A_d = \left. \frac{\partial g}{\partial x^e} \right|_{\bar{q}}, \quad B_d = \left. \frac{\partial g}{\partial u} \right|_{\bar{q}}, \quad E_d = \left. \frac{\partial g}{\partial \theta} \right|_{\bar{q}}, \quad (39)$$

using the implemented one-sided state perturbations and centered input and grade perturbations. Thus, no additional continuous-to-discrete conversion is applied.

## 2) Offline LPV Database

The scheduling grid is formed over speed, yaw rate, and grade. At each grid point, the nominal longitudinal force is

$$F_{cmd}^* = c_r mg \cos \theta^* + \frac{1}{2} \rho_{air} C_D A (v^*)^2 + mg \sin \theta^*, \quad (40)$$

$$u^* = [F_{cmd}^* \quad \omega^*]^T.$$

The database uses 11 speed points from 0.02 to 1.20 m/s, 15 yaw-rate points from  $-1.2$  to  $1.2$  rad/s, and 25 grade points from  $-12^\circ$  to  $12^\circ$ . It therefore stores  $11 \times 15 \times 25 = 4125$  local models, each containing  $(A_d, B_d, C_d, D_d, E_d)$ , its operating point, and numerical metadata. Online model updating is consequently reduced to interpolation rather than repeated nonlinear linearization.

## 3) Synchronized Scheduling and Interpolation

The same conditioned scalar is supplied to every grade-dependent controller path,

$$\begin{aligned} \mathbf{q}_k &= [v_k \quad \omega_k \quad \theta_k^{sch}]^T, \\ d_k &= \theta_k^{sch}, \\ F_{eq,k} &= mg (\sin \theta_k^{sch} + c_r \cos \theta_k^{sch}). \end{aligned} \quad (41)$$

This single-source rule prevents the LPV matrices, measured-disturbance channel, and grade feedforward from receiving differently filtered versions of the grade. The tracking weights, input weights, increment weights, and input envelopes are updated from the same operating condition.

After  $\mathbf{q}_k$  is saturated to the database limits, trilinear interpolation is performed within the enclosing cell. For any stored matrix  $M \in \{A_d, B_d, C_d, D_d, E_d\}$ ,

$$\begin{aligned} \xi_r &= \frac{\mathbf{q}_{k,r} - \mathbf{q}_r^-}{\mathbf{q}_r^+ - \mathbf{q}_r^-}, \\ \lambda_b &= \prod_{r=1}^3 \xi_r^{b_r} (1 - \xi_r)^{1-b_r}, \\ M(\mathbf{q}_k) &= \sum_{b \in \{0,1\}^3} \lambda_b M_b, \quad \sum_b \lambda_b = 1. \end{aligned} \quad (42)$$

The interpolated model is held fixed within the current prediction horizon. Smooth condition-dependent maps use  $h(z) = 3z^2 - 2z^3$  before scaling configured weights and bounds, avoiding abrupt changes at scheduling-cell boundaries.

## 4) Constrained MPC Optimization

No road-grade preview is assumed. The current conditioned estimate is held over the prediction horizon,  $\theta_{k+i|k} = \theta_k^{sch}$  for  $i = 0, \dots, N_p$ , and the corresponding equilibrium command is updated at every sampling instant. The grade disturbance is included in the lifted deviation prediction through

$$\mathbf{v}_k = \begin{bmatrix} \tilde{u}_k \\ \tilde{\theta}_k \end{bmatrix}, \quad \bar{B}_k = [B_d(\mathbf{q}_k) \quad E_d(\mathbf{q}_k)], \quad (43)$$

which yields

$$Y_k = F_k \tilde{x}_k^e + G_k \tilde{U}_k + H_k \tilde{\Theta}_k. \quad (44)$$

The scheduled feedforward term is used to form the input operating point,

$$\bar{u}_k = [F_{eq,k} + \frac{1}{2} \rho_{air} C_D A v_k^2 \quad \omega_k^{ref}]^T. \quad (45)$$

Thus,  $F_{eq,k}$  is not added again after optimization: the optimizer returns the absolute command  $u_{k|k} = \bar{u}_k + \tilde{u}_{k|k}$ , and only that command is applied to the plant. The finite-horizon objective is

$$\begin{aligned} J_k &= \sum_{i=1}^{N_p} \|y_{k+i|k} - y_{k+i}^{ref}\|_{Q_k}^2 \\ &+ \sum_{i=0}^{N_c-1} \|\tilde{u}_{k+i|k}\|_{R_k}^2 + \sum_{i=0}^{N_c-1} \|\Delta u_{k+i|k}\|_{R_{\Delta,k}}^2 \\ &+ w_{ECR} \|\epsilon_k\|_2^2, \end{aligned} \quad (46)$$

subject to

$$u_{\min,k} \leq u_{k+i|k} \leq u_{\max,k}, \quad \Delta u_{\min} \leq \Delta u_{k+i|k} \leq \Delta u_{\max}. \quad (47)$$

where  $\Delta u_{k+i|k} = u_{k+i|k} - u_{k+i-1|k}$ , with  $u_{k-1|k}$  equal to the previously applied command. The lateral and heading errors use the soft bounds

$$y_{\min} - \epsilon_k \leq y_{k+i|k} \leq y_{\max} + \epsilon_k, \quad \epsilon_k \geq 0, \quad (48)$$

and the runtime slack penalty  $w_{ECR}$  is held fixed across controller variants.

The controller structure, nominal values, interpolation maps, admissible ranges, and condition-dependent update laws in Table 4 were fixed before comparison. The realized  $Q_k$ ,  $R_k$ ,  $R_{\Delta,k}$ , and input envelopes can nevertheless vary online because the grade-source signal is the deliberately varied scheduler input and the same fixed update law is applied in every case. Accordingly, a fixed controller configuration means the absence of case-specific retuning, not equality of instantaneous scheduled parameters.

## 5) Online Receding-Horizon Execution

At every sampling instant, the deployed loop executes the following sequence:

- 1) acquire the current AGV responses and form the 22-D feature vector;
- 2) normalize and augment the features, update the historical window, and evaluate the ModernTCN estimator;
- 3) execute Algorithm 1 to update the Qualified observer, fuse the two estimates, and obtain the conditioned grade  $\theta_k^{sch}$ ;
- 4) use the same  $\theta_k^{sch}$  to form the scheduling vector, measured-disturbance input, grade feedforward, weights, and bounds;
- 5) interpolate  $(A_d, B_d, C_d, D_d, E_d)$  and solve the constrained quadratic program with  $N_p = 150$  and  $N_c = 30$ ; and

**TABLE 4. Implemented LPV-MPC settings and admissible online ranges used in closed-loop simulation.**

| Item                     | Symbol                | Value                                                                      | Unit / Description                        |
|--------------------------|-----------------------|----------------------------------------------------------------------------|-------------------------------------------|
| Sampling time            | $T_s$                 | 0.0100                                                                     | s                                         |
| Prediction horizon       | $N_p$                 | 150                                                                        | 1.50 s                                    |
| Control horizon          | $N_c$                 | 30                                                                         | 0.30 s                                    |
| Nominal tracking weight  | $Q_0$                 | diag(100, 100, 15, 3)                                                      | $e_y, e_\psi, e_v, e_\omega^{\text{trk}}$ |
| Nominal input weight     | $R_0$                 | diag( $3 \times 10^{-5}$ , $3 \times 10^{-5}$ )                            | $F_{cmd}, \omega_{cmd}$                   |
| Nominal increment weight | $R_{\Delta,0}$        | diag( $10^{-3}$ , $10^{-3}$ )                                              | $\Delta F_{cmd}, \Delta \omega_{cmd}$     |
| Online range             | $Q_k$                 | diag(50, 50, 7.5, 1.5) to diag(150, 150, 22.5, 4.5)                        | before condition-dependent gains          |
| Online range             | $R_k$                 | diag( $1.5, 1.5$ ) $\times 10^{-5}$ to diag( $4.5, 4.5$ ) $\times 10^{-5}$ | scheduled input penalty                   |
| Online range             | $R_{\Delta,k}$        | diag( $0.5, 0.5$ ) $\times 10^{-3}$ to diag( $1.5, 1.5$ ) $\times 10^{-3}$ | scheduled increment penalty               |
| Input constraint         | $F_{cmd}$             | [−600, 600]                                                                | N; online envelope [−720, 720]            |
| Input constraint         | $\omega_{cmd}$        | [−1.2, 1.2]                                                                | rad/s; online envelope [−1.44, 1.44]      |
| Increment constraint     | $\Delta F_{cmd}$      | [−400, 400]                                                                | N/step                                    |
| Increment constraint     | $\Delta \omega_{cmd}$ | [−0.9, 0.9]                                                                | (rad/s)/step                              |
| Soft output constraint   | $e_y$                 | [−1.0, 1.0]                                                                | m                                         |
| Soft output constraint   | $e_\psi$              | [−0.5, 0.5]                                                                | rad                                       |
| Runtime slack penalty    | $w_{\text{ECR}}$      | $10^4$                                                                     | external Adaptive MPC weight              |
| Implementation           | Solver                | MATLAB MPC Toolbox                                                         | Adaptive MPC with online model update     |

- 6) apply only the first optimized command to the nonlinear plant and repeat the sequence at the next sample.

This ordered implementation preserves a traceable interface between temporal perception and model-based control while keeping the plant, prediction model, and disturbance update synchronized.

#### IV. EXPERIMENTAL SETUP AND EVALUATION PROTOCOL

##### A. RESEARCH QUESTIONS AND EVALUATION DESIGN

The evaluation was organized around five sequential research questions. RQ1 asked whether road-grade scheduling materially affects the closed loop of the LPV-MPC with fixed tuning. RQ2 asked whether ModernTCN-delta improves offline grade estimation relative to ModernTCN-22D, GRU-22D, and TCN-22D under a common data protocol. RQ3 asked whether the offline estimator advantage remained visible after deployment in the common closed-loop simulation interface. RQ4 examined how the Qualified observer complements the history-based estimate through the bounded fusion interface. RQ5 quantified the resulting closed-loop changes relative to MTCN-LPV-MPC. A supplementary analysis compared the core inference cost of the four temporal estimators. This ordering separates the value of the scheduling variable from estimator selection, fusion behavior, controller performance, and estimator-level computation.

The estimator and controller comparisons used fixed data, models, routes, and numerical settings within each evidence family. The fusion study used one locked interface configuration and 60 matched route–seed pairs. Source-level comparisons were kept separate from paired closed-loop effects, and runtime integrity was checked independently of performance aggregation.

##### B. SIMULATION PLATFORM AND FIXED CONTROLLER CONFIGURATION

All closed-loop cases used the same revision of the nonlinear AGV plant model and the same LPV-MPC implementation. The controller sample period was 0.01 s, the prediction horizon

was 150 steps (1.5 s), and the control horizon was 30 steps (0.3 s). Controller structure, nominal weights, scheduling maps and ranges, increment and output bounds, soft-constraint weights, LPV database, and controller cache were fixed before any comparison was run. The implemented controller settings are listed in Table 4. The realized scheduled weights and input envelopes were allowed to vary through the same fixed update law because grade source was the experimental factor; no case-specific retuning was performed. Table 5 summarizes the data, estimator, route, and comparison configuration held fixed across the reported evidence families.

##### C. SIMULATION DATA GENERATION AND DATASET CONSTRUCTION

The dataset generator used master seed 20260511 to produce 102 continuous simulation runs from 51 route templates, with two random realizations for each template. Windowing was applied only after the run-level split, yielding 16529 training, 3695 validation, and 3602 test windows. The target corresponded to the current window end. Only causal filtering reproducible in online operation was allowed, and normalization statistics were fitted to the training runs before application to validation, test, and online inputs. None of the six closed-loop route files or their recorded samples was used to form these windows.

The base input contains 22 proprioceptive and dynamics-derived channels. ModernTCN-delta augments each normalized vector with first differences at lags 1, 2, and 4, producing 88 channels while retaining the same 128-step history. It was trained with AdamW for at most 120 epochs using batch size 256, learning rate  $10^{-3}$ , weight decay  $10^{-4}$ , early-stopping patience 25, and a minimum of 30 epochs; the retained checkpoint followed a pre-specified validation rule for each of the ten seeds. Algorithm 2 summarizes the leakage-controlled construction order and makes explicit that run assignment precedes window extraction and that true grade is retained only as an endpoint label.

**TABLE 5. Data, estimator, and comparison configuration held fixed across all reported evidence families.**

| Item                  | Fixed value                                                                                            | Execution meaning                                                                                                                                                 |
|-----------------------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Plant and sampling    | fixed plant revision (v2); $T_s = 0.01$ s                                                              | Identical nonlinear plant and sampling interface for every controller case                                                                                        |
| Dataset               | 588102 raw samples; 102 continuous runs                                                                | 51 route templates with two realizations per template; generator seed 20260511                                                                                    |
| Reference speed       | Training templates: 0.80–1.18 m/s (sample-weighted mean, 0.992 m/s); closed-loop routes: 0.82–0.93 m/s | Separates the data-generation coverage from the lower-speed closed-loop benchmark                                                                                 |
| History and input     | 128 steps (1.28 s); 22D or 88D                                                                         | The 88D input is $[x_t, \Delta_1 x_t, \Delta_2 x_t, \Delta_4 x_t]$ after train-set normalization; deltas are clipped at 5                                         |
| Run split             | 71/15/16 train/validation/test runs                                                                    | Run-disjoint; scalars fitted on training runs only; no closed-loop route file or sample was reused                                                                |
| Closed-loop benchmark | Six predefined routes; ten learned-model seeds; 60 matched MTCN–Fusion route–seed pairs                | Representative traces explain runtime behavior; quantitative conclusions use the complete matched sets                                                            |
| Comparison control    | Controller settings in Table 4; one fixed interface configuration within each comparison family        | Same plant, initialization, LPV database, update laws, and no case-specific retuning                                                                              |
| Primary estimator     | ModernTCN-delta, seeds 1, 7, 11, 21, 42, 73, 101, 202, 340, 520                                        | Seed 42 was pre-specified for controller and fusion traces; the four-estimator P2 trace used representative seed 73; all ten seeds entered quantitative summaries |

### Algorithm 2 Leakage-controlled construction of run-disjoint 22D and 88D datasets

**Require:** 51 route templates  $\mathcal{T}$ ; two fixed realizations per template; causal 22D extractor;  $L = 128$ ;  $\mathcal{L} = \{1, 2, 4\}$

**Ensure:** run-disjoint paired 22D and ModernTCN-delta 88D window sets  $\mathcal{D}_{tr}$ ,  $\mathcal{D}_{va}$ , and  $\mathcal{D}_{te}$

```

1:  $\mathcal{R} \leftarrow \emptyset$ 
2: for each template  $\tau \in \mathcal{T}$  and realization  $r \in \{1, 2\}$  do
3:   Record one continuous run with causal 22D responses, endpoint grade targets, and run/template identifiers; append it to  $\mathcal{R}$ 
4: end for
5: Verify  $|\mathcal{R}| = 102$  and 588102 total raw samples
6: Split run identifiers 71/15/16 before windowing run-disjoint split
7: Fit  $\mu_{tr}$  and  $\sigma_{tr}$  using training-run 22D samples only
8: for each split  $s$  and each continuous run  $R \in s$  do
9:   Normalize  $R$  using  $\mu_{tr}$  and  $\sigma_{tr}$ 
10:  Form edge-padded,  $\pm 5$ -clipped  $\Delta_\ell \mathbf{z}_k$  for  $\ell \in \mathcal{L}$  and concatenate  $\tilde{\mathbf{z}}_k = [\mathbf{z}_k, \Delta_1 \mathbf{z}_k, \Delta_2 \mathbf{z}_k, \Delta_4 \mathbf{z}_k] \in \mathbb{R}^{88}$ 
11:  Extract the endpoint-aligned 128-step 22D/88D windows wholly within  $R$ ; attach endpoint grade labels and append to  $\mathcal{D}_s$ 
12: end for
13: Verify  $|\mathcal{D}_{tr}|/|\mathcal{D}_{va}|/|\mathcal{D}_{te}| = 16529/3695/3602$ 
14: return  $\mathcal{D}_{tr}$ ,  $\mathcal{D}_{va}$ , and  $\mathcal{D}_{te}$ 

```

### D. COMPARISON METHODS AND GRADE-SOURCE CONTROLS

The comparison protocol separated estimator architecture, input representation, scheduling necessity, and fusion value. ModernTCN-22D, GRU-22D, and TCN-22D shared the 22D input and common data protocol, whereas ModernTCN-delta was paired with ModernTCN-22D to isolate the contribution of the multi-scale delta bank. Zero-grade and truth-driven conditioned scheduling established the control relevance of grade information. The fusion study then compared MTCN-LPV-MPC with a matched counterpart in which the Qualified observer supplied only a bounded complementary correction. ModernTCN-delta remained the primary grade source, and the truth-driven signal remained an analysis-only ref-

erence that was not read by the deployed fusion interface. Across controller comparisons, the plant, initialization, LPV database, controller structure, nominal settings, scheduling maps, admissible ranges, and update laws were held fixed without case-specific retuning.

### E. CLOSED-LOOP ROUTES, SEEDS, AND STATISTICAL UNITS

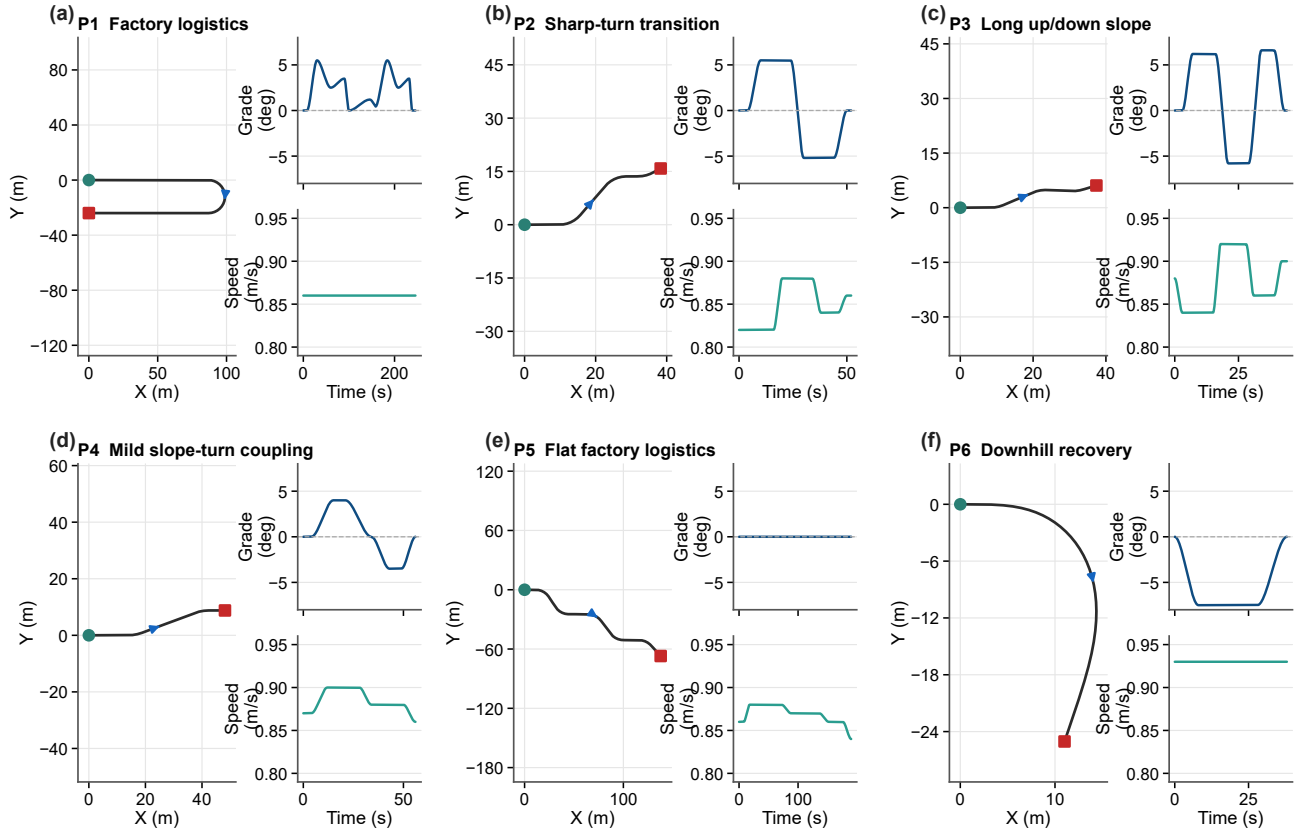
The closed-loop evaluation used six predefined routes: P1 Factory logistics, P2 Sharp-turn transition, P3 Long up/down slope, P4 Mild slope-turn coupling, P5 Flat factory logistics, and P6 Downhill recovery. Zero-grade and truth-driven scheduling contributed one deterministic case per route, whereas the learned-source comparisons contributed ten model-seed cases per route. The fusion analysis therefore preserved the model-seed structure and comprised 60 matched MTCN–Fusion route–seed pairs. Figure 3 shows the six evaluation routes.

The selected fusion configuration was applied uniformly to the same six routes and ten model seeds used by MTCN-LPV-MPC. The resulting 60 matched pairs were aggregated with the hierarchical seed-first bootstrap described below. All quantitative fusion results use the complete matched set, while the route- and seed-level records are retained as supplementary data.

Model seed 42 was retained for the controller and fusion visualizations specified by those analyses. The four-estimator trace in Fig. 6 uses seed 73 under a separate representativeness rule. Quantitative conclusions use all ten seeds and all six routes, while the representative traces illustrate runtime behavior alongside the complete summaries.

### F. METRICS AND STATISTICAL ANALYSIS

Offline primary metrics were grade mean absolute error (MAE) and either the pre-specified  $|\theta| \leq 10^\circ$  P95 absolute error or the edge-region P95 absolute error, depending on the comparison. The frozen edge-region metric was the larger of the P95 absolute errors evaluated over  $-10^\circ \leq \theta \leq -8^\circ$  and  $8^\circ \leq \theta \leq 10^\circ$ . Closed-loop estimator accuracy was summarized by route-wise grade MAE after the common 0.5-s warm-up and by the complete-trace P95 absolute error. The



**FIGURE 3.** Six-route closed-loop benchmark. Each panel shows the complete planar reference path together with the aligned grade and reference-speed profiles. Green circles, red squares, and blue arrowheads mark route starts, route ends, and travel direction, respectively.

three inferential primary closed-loop outcomes were lateral-error RMS, heading-error RMS, and the composite input-increment index  $J_{\Delta u}$ . Lower values favor the target method. The composite input-increment index is defined as a control-regularity summary rather than a physical energy measure.

Paired effects are reported as comparator minus target so that positive values favor the target. Offline intervals resampled paired model seeds. Learned closed-loop intervals resampled model seeds first and then retained all six matched routes within each resampled seed; deterministic comparisons resampled routes. All reported bootstrap intervals used 10000 iterations and seed 20260715. Table 6 records the unit and interpretation of each evidence family.

### G. REPRODUCIBILITY AND REPORTING BOUNDARIES

The dataset, feature specification, model checkpoints, plant revision, six route definitions, LPV database, controller cache, metrics, thresholds, and seed set were fixed and identified by file checksums. A completeness check confirmed 40/40 offline estimator rows, 240/240 estimator closed-loop rows, 18/18 scheduling-necessity cases, and 60/60 matched fusion cases. A source cross-check covered the reported paired intervals and matched-case summaries without discrepancy.

## V. RESULTS AND DISCUSSION

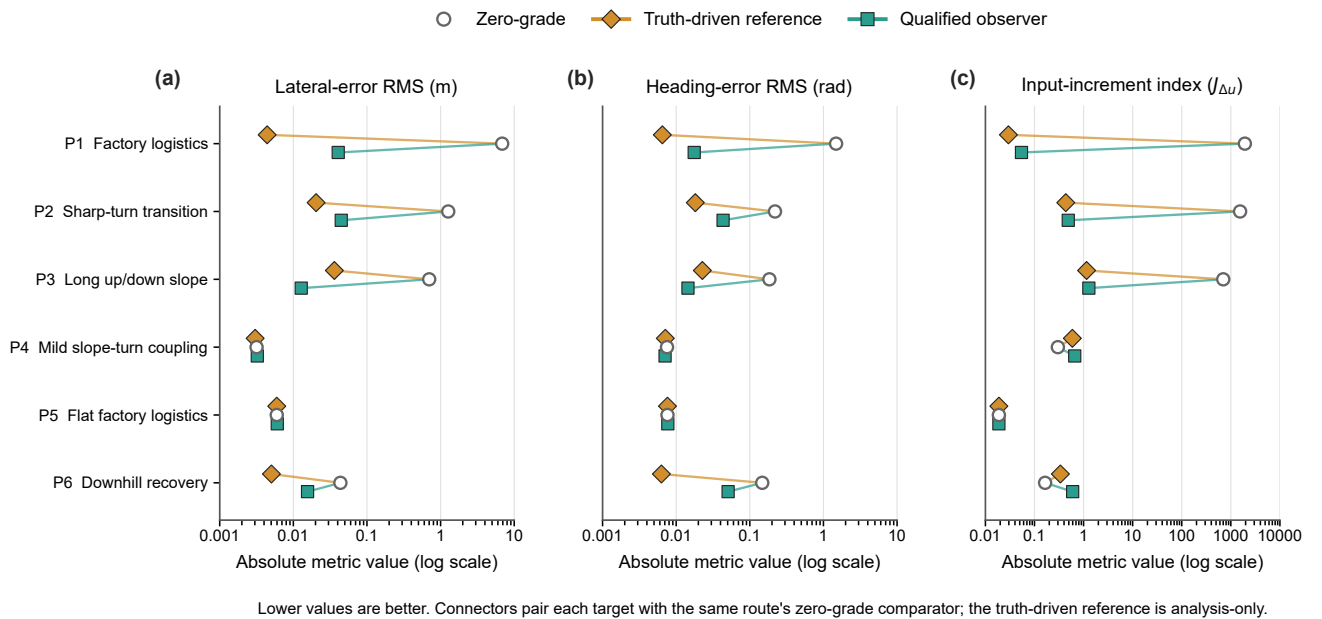
### A. RQ1: DOES ACCURATE GRADE SCHEDULING AFFECT THE LPV-MPC CLOSED LOOP?

Across the six routes, the absolute means followed the expected aggregate ordering from missing, through estimated, to exact grade information. Zero-grade scheduling yielded lateral-error RMS, heading-error RMS, and  $J_{\Delta u}$  means of 1.48046 m, 0.34204 rad, and 696.014, respectively. The Qualified observer reduced these values to 0.02051 m, 0.02347 rad, and 0.51646, while the analysis-only truth-driven reference yielded the lowest corresponding means of 0.01248 m, 0.01139 rad, and 0.42684. Relative to zero-grade scheduling, the mean reductions under the truth-driven reference were 1.46798 m, 0.33065 rad, and 695.587; the corresponding reductions under the Qualified observer were 1.45995 m, 0.31857 rad, and 695.497. All six bootstrap confidence intervals reported in Table 7 excluded zero. These aggregate results establish the control relevance of grade scheduling and indicate the additional performance potential provided by a more accurate scheduling signal.

The route-resolved results in Fig. 4 support a clear aggregate hierarchy from missing, through estimated, to exact grade information. The truth-driven condition provides an analysis-only upper reference, whereas standalone scheduling with the Qualified observer shows that a causal grade estimate recovers most of the closed-loop benefit. RQ1 therefore establishes that grade scheduling materially improves the

**TABLE 6.** Evaluation metrics, statistical units, and interpretation rules.

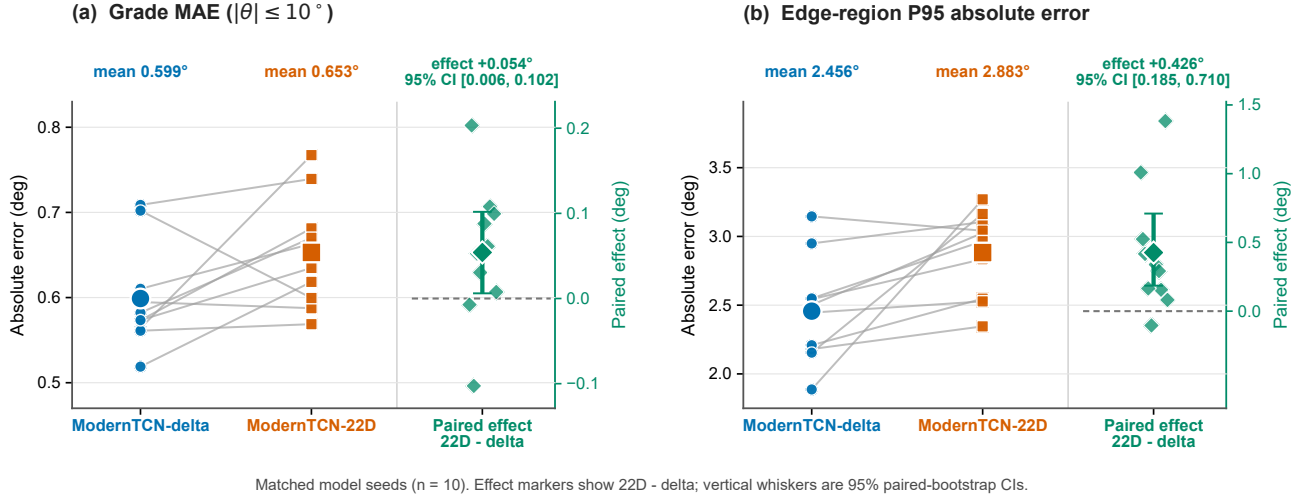
| Evidence family               | Reported quantity                                                                                | Unit      | Statistical unit                                   | Decision rule                                                        |
|-------------------------------|--------------------------------------------------------------------------------------------------|-----------|----------------------------------------------------|----------------------------------------------------------------------|
| Offline grade estimation      | MAE; pre-specified P95 absolute error; edge-region P95 absolute error                            | deg       | Ten paired model seeds                             | Positive comparator-minus-target effect favors target                |
| Closed-loop grade estimation  | Route-wise grade MAE; complete-trace P95 absolute error                                          | deg       | Ten model seeds per route                          | Descriptive mean $\pm$ SD; lower values favor the method             |
| Primary closed-loop inference | $e_y$ RMS; $e_\psi$ RMS; $J_{\Delta u}$                                                          | m; rad; – | Six paired routes or ten seeds $\times$ six routes | Route or hierarchical seed-first bootstrap                           |
| Fusion configuration          | Fixed gain, innovation, NIS, correction, rate, and scheduling guards                             | –         | One locked interface configuration                 | Applied identically to all matched closed-loop cases                 |
| Fusion closed-loop comparison | $e_y$ RMS; $e_\psi$ RMS; $J_{\Delta u}$                                                          | m; rad; – | Ten seeds $\times$ six matched routes              | Hierarchical seed-first paired bootstrap and matched mean reductions |
| Execution integrity           | Case completion, constraint compliance, solver status, fallback identity, and runtime signal use | status    | 60 matched cases                                   | Checked independently of the performance aggregation                 |
| Estimator core runtime        | p50, p95, p99, > 10-ms count                                                                     | ms        | 10000 timed calls after 500 warm-ups               | Compare methods only within the common ONNX Runtime CPU backend      |

**FIGURE 4.** Route-resolved absolute closed-loop metrics under zero-grade scheduling, the truth-driven conditioned reference, and Qualified observer scheduling. Panels show (a) lateral-error RMS, (b) heading-error RMS, and (c) the composite input-increment index. White circles denote zero-grade scheduling, orange diamonds the truth-driven reference, and teal squares the Qualified observer; connectors pair each grade-informed result with the same route's zero-grade comparator. Lower values indicate better performance, and logarithmic axes accommodate route-to-route scale differences. The truth-driven conditioned reference is analysis-only and represents an idealized scheduling reference rather than a deployable grade source.**TABLE 7.** Six-route paired effects relative to zero-grade LPV-MPC. Effects are zero-grade minus target, so positive values favor the target. Boldface mean effects have 95% confidence intervals excluding zero. The truth-driven conditioned reference is analysis-only.

| Target source                      | Metric             | Mean effect    | 95% CI              | Routes |
|------------------------------------|--------------------|----------------|---------------------|--------|
| Truth-driven conditioned reference | $e_y$ RMS (m)      | <b>1.46798</b> | [0.11715, 3.74682]  | 6      |
| Truth-driven conditioned reference | $e_\psi$ RMS (rad) | <b>0.33065</b> | [0.05065, 0.80004]  | 6      |
| Truth-driven conditioned reference | $J_{\Delta u}$     | <b>695.587</b> | [116.408, 1339.454] | 6      |
| Qualified observer                 | $e_y$ RMS (m)      | <b>1.45995</b> | [0.11917, 3.72848]  | 6      |
| Qualified observer                 | $e_\psi$ RMS (rad) | <b>0.31857</b> | [0.04460, 0.79161]  | 6      |
| Qualified observer                 | $J_{\Delta u}$     | <b>695.497</b> | [116.277, 1339.412] | 6      |

LPV-MPC closed loop and that higher scheduling accuracy provides further performance potential. The standalone observer condition is included only to isolate the value of grade quality; the deployed fusion architecture is evaluated separately in RQ4 and RQ5. Table 7 provides the corresponding

paired effects and route-bootstrap confidence intervals.



**FIGURE 5.** Matched-seed offline comparison of ModernTCN input representations. (a) Grade MAE over test windows with  $|\theta| \leq 10^\circ$ . (b) Edge-region P95 absolute grade error. Blue circles and vermillion squares show individual results for ModernTCN-delta and ModernTCN-22D, respectively; gray lines connect matched model seeds, and larger markers denote arithmetic means. Green diamonds show seed-paired effects defined as ModernTCN-22D minus ModernTCN-delta. The larger diamond and vertical whiskers denote the mean effect and its 95% paired-seed bootstrap confidence interval. Positive effects favor ModernTCN-delta ( $n = 10$ ).

### B. RQ2: DOES THE MULTI-SCALE DELTA BANK IMPROVE OFFLINE GRADE ESTIMATION?

Figure 5 isolates the effect of the multi-scale delta bank using ten matched model seeds. Over test windows with  $|\theta| \leq 10^\circ$ , the mean grade MAE decreased from  $0.65300^\circ$  for ModernTCN-22D to  $0.59888^\circ$  for ModernTCN-delta. The paired effect was  $0.05412^\circ$  with a 95% confidence interval of  $[0.00608, 0.10180]$ . In the edge region, the mean P95 absolute error decreased from  $2.88256^\circ$  to  $2.45649^\circ$ , corresponding to a paired effect of  $0.42608^\circ$   $[0.18517, 0.70987]$ .

The broader four-estimator benchmark is reported in Table 8. ModernTCN-delta achieved the lowest mean MAE and pre-specified  $|\theta| \leq 10^\circ$  P95 absolute error, with supported paired advantages over GRU-22D and TCN-22D.

Figure 5 shows all ten matched seed pairs for the two ModernTCN input representations, while Table 8 retains the complete four-estimator comparison. Together, they identify ModernTCN-delta as the strongest of the four tested configurations under the present evaluation protocol.

### C. RQ3: DOES THE OFFLINE ESTIMATOR ADVANTAGE PERSIST IN CLOSED-LOOP GRADE ESTIMATION?

To determine whether the offline estimator ordering remained visible after deployment in the common simulation interface, the four-estimator closed-loop grade traces were compared. Figure 6 shows the complete P2 Sharp-turn transition route. Model seed 73 was selected by a representativeness rule and ranked fourth among the ten P2 seeds by ModernTCN-delta case-level MAE. The complete six-path, ten-seed summary is reported in Table 9.

In the displayed case, ModernTCN-delta achieved the lowest MAE ( $0.504^\circ$ ) and P95 absolute error ( $1.138^\circ$ ), while the four trajectories reveal transition and steady-state char-

acteristics beyond the offline distributions in Fig. 5. Across the complete closed-loop benchmark, ModernTCN-delta also had the lowest all-path mean MAE ( $0.423^\circ$ ) and P95 absolute error ( $1.237^\circ$ ). It ranked first on four of six routes for both metrics; ModernTCN-22D led P5, while GRU-22D and TCN-22D provided the lowest P3 MAE and P95, respectively.

The offline and closed-loop estimator results therefore answer complementary questions: RQ2 isolates the delta-bank representation under a common data protocol, whereas the evidence above describes its behavior after integration into the closed-loop simulation. Controller-level benefit is evaluated separately in RQ5, where the scheduling interface, route geometry, constrained optimizer, and command regularization determine how grade-estimation errors are translated into tracking metrics.

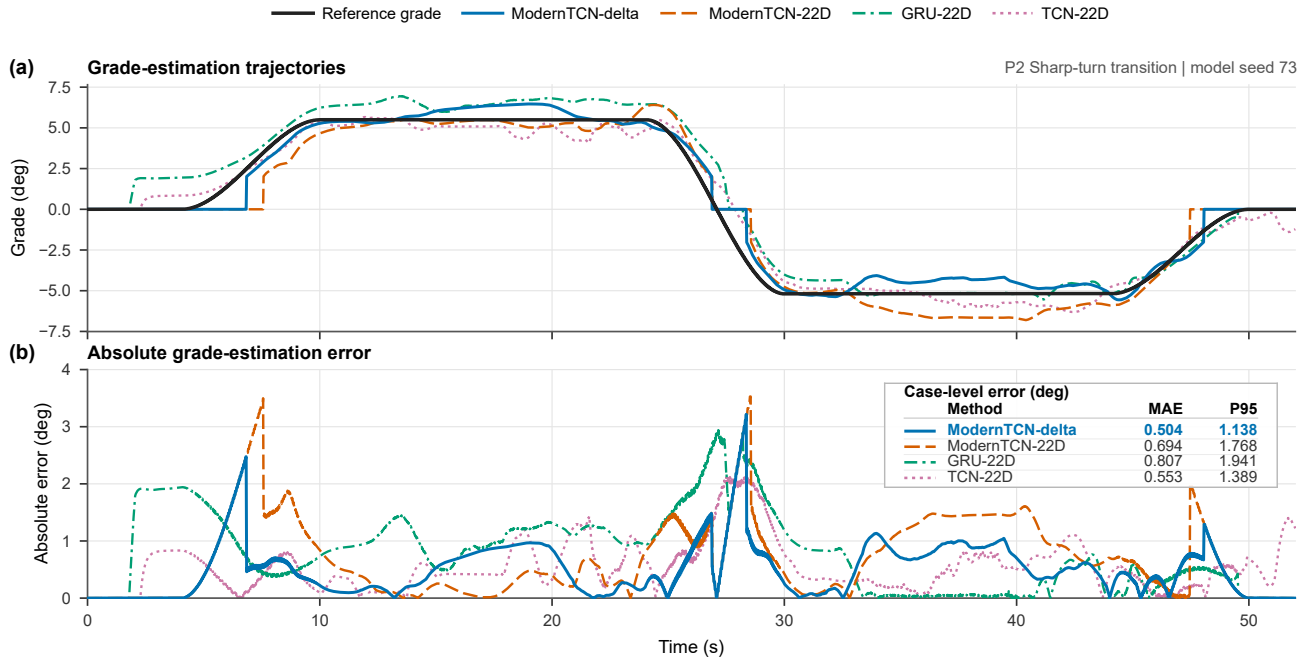
The final closed-loop MTCN-LPV-MPC benchmark achieved mean lateral-error RMS of  $0.02635$  m, heading-error RMS of  $0.02216$  rad, and  $J_{\Delta u} = 0.93053$  over six routes and ten model seeds. These values define the matched controller baseline used in RQ5. The fusion comparison retains the same route-seed structure so that the effect of the bounded correction supplied by the Qualified observer is evaluated directly rather than inferred from a separate deterministic source.

### D. RQ4: HOW DOES THE QUALIFIED OBSERVER COMPLEMENT MODERNTCN?

The Qualified observer uses the six-axis inertial configuration described in Section III-D and contributes through the bounded correction channel. ModernTCN-delta remains the primary grade source, while the observer contributes when calibrated source uncertainty, estimated helpfulness, and innovation consistency jointly support the correction. A

**TABLE 8. Offline estimator comparison over ten matched model seeds. Panel A reports absolute means. Panel B reports comparator-minus-ModernTCN-delta paired effects; positive effects favor ModernTCN-delta, and boldface intervals exclude zero.**

| Panel A: Absolute offline means                          |                                   |             |                                   |                          |             |
|----------------------------------------------------------|-----------------------------------|-------------|-----------------------------------|--------------------------|-------------|
| Method                                                   | MAE mean (deg)                    | MAE support | P95 mean (deg)                    | P95 region               | P95 support |
| ModernTCN-delta                                          | 0.59888                           | —           | 1.67777                           | $ \theta  \leq 10^\circ$ | —           |
| ModernTCN-22D                                            | 0.65300                           | —           | 1.80626                           | $ \theta  \leq 10^\circ$ | —           |
| GRU-22D                                                  | 0.88583                           | —           | 2.25950                           | $ \theta  \leq 10^\circ$ | —           |
| TCN-22D                                                  | 0.91537                           | —           | 2.63580                           | $ \theta  \leq 10^\circ$ | —           |
| Panel B: Paired comparator-minus-ModernTCN-delta effects |                                   |             |                                   |                          |             |
| Comparator                                               | MAE effect [95% CI] (deg)         | MAE support | P95 effect [95% CI] (deg)         | P95 region               | P95 support |
| ModernTCN-22D                                            | <b>0.05412 [0.00608, 0.10180]</b> | 8/10        | <b>0.42608 [0.18517, 0.70987]</b> | Edge region              | 9/10        |
| GRU-22D                                                  | <b>0.28694 [0.18468, 0.37271]</b> | 9/10        | <b>0.58172 [0.38462, 0.80045]</b> | $ \theta  \leq 10^\circ$ | 10/10       |
| TCN-22D                                                  | <b>0.31648 [0.24668, 0.38012]</b> | 10/10       | <b>0.95802 [0.69170, 1.25456]</b> | $ \theta  \leq 10^\circ$ | 10/10       |



**FIGURE 6. Representative closed-loop grade-estimation results on the P2 Sharp-turn transition route with model seed 73. (a) Reference grade and estimates produced by ModernTCN-delta, ModernTCN-22D, GRU-22D, and TCN-22D over the complete route. (b) Corresponding absolute grade-estimation errors. The inset reports case-level MAE after the common 0.5-s warm-up and complete-trace P95 absolute error. The displayed seed is a representative case selected independently of minimum error; the complete six-path, ten-seed results are reported in Table 9.**

common set of activation, innovation-consistency, correction-amplitude, rate, and scheduling guards was fixed for all matched cases, as summarized in Table 10. This design retains the learned estimate as the nominal signal and conditions the observer contribution before the fused grade enters the scheduling filter.

Figure 7 illustrates how these conditions act jointly at runtime. P1 was retained for the qualitative trace because it contains grade variation, turning, and speed changes in one route, while model seed 42 was specified a priori for qualitative visualization. Within this case, the 129.95–144.94 s interval was chosen by a deterministic maximum-active-fraction rule over fixed 15-s windows. The fused estimate therefore coincides exactly with ModernTCN-delta during fallback and receives a limited observer correction in eligible intervals. This trace explains the runtime mechanism, while quantitative inference uses all 60 matched route-seed pairs.

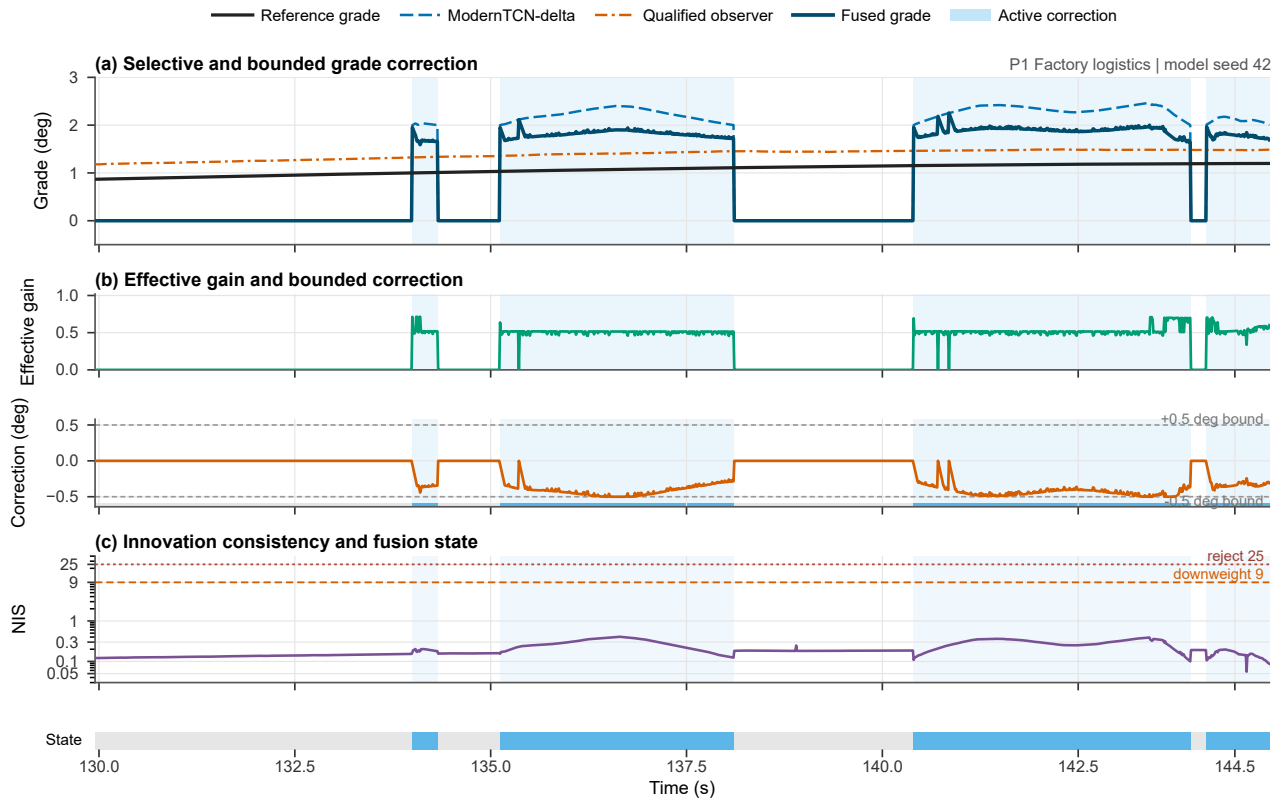
The fixed interface was evaluated over all 60 matched route-seed cases, with ground-truth grade excluded from the runtime fusion inputs. All cases completed the prescribed constraint, solver, timeout, diagnostic, truth-access, and fallback-identity checks. Table 10 summarizes the configuration and execution checks, while RQ5 reports the paired closed-loop performance effects.

#### E. RQ5: DOES FUSION ADD VALUE BEYOND MODERNTCN IN CLOSED LOOP?

Figure 8 and Table 11 jointly report the matched closed-loop comparison between MTCN-LPV-MPC and Fusion-LPV-MPC. Figure 8 shows the complete route-seed distribution and route-level heterogeneity, whereas Table 11 gives the aggregate means, relative reductions, and paired bootstrap intervals. Across the 60 route-seed pairs, the mean lateral-error RMS decreased from 0.02635 m to 0.01891 m,

**TABLE 9. Path-wise closed-loop grade-estimation accuracy over ten matched model seeds. Panel A reports grade MAE after the common 0.5-s warm-up, and Panel B reports the complete-trace P95 absolute error. Entries are mean  $\pm$  SD over ten seeds for each path. The All paths row first averages the six routes within each seed and then summarizes the ten seed-level means. Lower values indicate better performance; the best mean in each row is highlighted.**

| Path                                     | ModernTCN-delta                     | ModernTCN-22D                       | GRU-22D                             | TCN-22D                             |
|------------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
| <i>Panel A: Grade MAE (deg)</i>          |                                     |                                     |                                     |                                     |
| P1 Factory logistics                     | <b>0.458 <math>\pm</math> 0.104</b> | 0.492 $\pm$ 0.156                   | 0.827 $\pm$ 0.408                   | 2.387 $\pm$ 0.833                   |
| P2 Sharp-turn transition                 | <b>0.593 <math>\pm</math> 0.107</b> | 0.645 $\pm$ 0.078                   | 0.698 $\pm$ 0.197                   | 0.783 $\pm$ 0.276                   |
| P3 Long up/down slope                    | 0.791 $\pm$ 0.121                   | 0.880 $\pm$ 0.087                   | <b>0.763 <math>\pm</math> 0.151</b> | 0.846 $\pm$ 0.232                   |
| P4 Mild slope-turn coupling              | <b>0.036 <math>\pm</math> 0.062</b> | 0.037 $\pm$ 0.064                   | 1.397 $\pm$ 0.451                   | 1.237 $\pm$ 0.614                   |
| P5 Flat factory logistics                | 0.087 $\pm$ 0.223                   | <b>0.046 <math>\pm</math> 0.092</b> | 1.206 $\pm$ 0.750                   | 1.249 $\pm$ 0.593                   |
| P6 Downhill recovery                     | <b>0.570 <math>\pm</math> 0.218</b> | 0.601 $\pm$ 0.193                   | 0.746 $\pm$ 0.227                   | 0.846 $\pm$ 0.517                   |
| All paths                                | <b>0.423 <math>\pm</math> 0.080</b> | 0.450 $\pm$ 0.046                   | 0.940 $\pm$ 0.267                   | 1.225 $\pm$ 0.434                   |
| <i>Panel B: P95 absolute error (deg)</i> |                                     |                                     |                                     |                                     |
| P1 Factory logistics                     | <b>1.434 <math>\pm</math> 0.434</b> | 1.451 $\pm$ 0.613                   | 2.528 $\pm$ 1.071                   | 10.888 $\pm$ 4.279                  |
| P2 Sharp-turn transition                 | <b>1.642 <math>\pm</math> 0.299</b> | 1.847 $\pm$ 0.310                   | 2.257 $\pm$ 0.558                   | 1.850 $\pm$ 0.453                   |
| P3 Long up/down slope                    | 2.289 $\pm$ 0.385                   | 2.812 $\pm$ 0.407                   | 2.270 $\pm$ 0.528                   | <b>2.197 <math>\pm</math> 0.495</b> |
| P4 Mild slope-turn coupling              | <b>0.229 <math>\pm</math> 0.723</b> | 0.244 $\pm$ 0.772                   | 2.353 $\pm$ 0.732                   | 1.469 $\pm$ 0.656                   |
| P5 Flat factory logistics                | 0.478 $\pm$ 1.024                   | <b>0.455 <math>\pm</math> 0.968</b> | 2.414 $\pm$ 0.641                   | 1.512 $\pm$ 0.688                   |
| P6 Downhill recovery                     | <b>1.351 <math>\pm</math> 0.249</b> | 1.418 $\pm$ 0.291                   | 1.792 $\pm$ 0.414                   | 2.000 $\pm$ 0.643                   |
| All paths                                | <b>1.237 <math>\pm</math> 0.357</b> | 1.371 $\pm$ 0.316                   | 2.269 $\pm$ 0.563                   | 3.319 $\pm$ 1.014                   |



**FIGURE 7. Representative operation of the bounded fusion interface on the P1 Factory logistics route with model seed 42. (a) Reference grade and the ModernTCN-delta, Qualified observer, and fused estimates. (b) Effective fusion gain and bounded grade correction. (c) Innovation consistency and active/fallback state. Shaded regions indicate active correction intervals. This representative case explains the runtime mechanism, while quantitative conclusions use all 60 matched route-seed pairs.**

corresponding to a 28.23% reduction. The paired MTCN-minus-Fusion effect was 0.00744 m with a 95% confidence interval of [0.00245, 0.01527]. The mean heading-error RMS decreased from 0.02216 rad to 0.01933 rad, corresponding to a 12.79% reduction, with a paired effect of 0.00283 rad and a 95% confidence interval of [0.00064, 0.00615].

The composite input-increment index decreased from 0.93053 to 0.67361, giving a 27.61% reduction in the matched mean. Its paired effect was 0.25692, with a 95% confidence interval of [−0.01349, 0.93333]. Together, the matched results show interval-supported reductions in the two primary tracking measures and a lower mean  $J_{\Delta u}$ . At the route-

**TABLE 10. Fixed fusion configuration and execution summary. Figure 7 shows the corresponding runtime behavior.**

| Component              | Fixed value                                                                                                   | Function in the deployed interface                                                            |
|------------------------|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Gain and activation    | $K_{\max} = 1.0$ ; helpfulness 0.6; slope entry/exit $0.5^\circ/0.3^\circ$                                    | Activates a bounded observer correction while retaining ModernTCN-delta as the primary source |
| Innovation consistency | Minimum innovation $0.25^\circ$ ; gate $2^\circ$ ; NIS thresholds 9/25                                        | Suppresses weak or inconsistent corrections before grade scheduling                           |
| Correction dynamics    | Magnitude bound $0.5^\circ$ ; rate bound $5^\circ \text{ s}^{-1}$                                             | Limits the amplitude and temporal variation of the complementary update                       |
| Scheduling guard       | Deadband $1.3^\circ$ ; range $[-10^\circ, 10^\circ]$                                                          | Produces the conditioned grade supplied consistently to LPV-MPC                               |
| Execution integrity    | 60/60 cases completed and passed the constraint, solver, timeout, diagnostic, and runtime truth-access checks | Confirms consistent execution and exact fallback identity with ModernTCN-delta                |
| Evidence scope         | 10 model seeds $\times$ 6 routes; one locked interface configuration                                          | Supports mechanism and execution claims; paired performance effects are evaluated in Table 11 |

aggregated level, P1 Factory logistics showed the clearest joint improvement, with lower mean lateral-error RMS, heading-error RMS, and  $J_{\Delta u}$  across the ten model seeds. The route-level markers in Fig. 8 show the complete dispersion of case-level effects around the aggregate improvements.

This matched comparison isolates the incremental effect of the fusion interface under the tested simulation conditions. The paired tracking intervals support the complementary use of the Qualified observer, while the input-increment index summarizes the corresponding change in control regularity. The bounded gain, correction limits, and exact fallback retain the history-based estimate as the nominal control-facing source and leave command generation in the model-based controller, consistent with hybrid physics–learning designs [36], [37].

#### F. WHAT IS THE CORE INFERENCE COST OF THE TESTED ESTIMATORS?

Under the common ONNX Runtime CPU backend, batch size one, one thread, 500 warm-ups, and 10000 timed calls, all four estimators had p95 core inference times below 1.2 ms (Table 12). ModernTCN-delta achieved a p95 of 0.4038 ms with no call above 10 ms. GRU-22D was the fastest core predictor, while TCN-22D produced one call above 10 ms. The sub-millisecond cores leave substantial margin within the 10-ms control period without requiring solver-side acceleration of the kind pursued by parallel NMPC implementations [38]. These same-backend measurements characterize estimator inference only and are not used to rank unmeasured wrappers or controller interfaces.

#### G. INTEGRATED DISCUSSION AND SCOPE

Taken together, the experiments establish a staged perception-to-control relationship. The truth-driven comparison first shows that grade is a consequential scheduling variable for the LPV-MPC with fixed tuning. ModernTCN-delta then improves grade estimation under the common offline protocol and retains the lowest all-path mean MAE and P95 error after deployment in the closed-loop simulation interface. These estimator-level results support grade-source selection, while controller performance is evaluated separately because scheduling, route geometry, constraints, and command regularization transform estimation errors before they appear in closed-loop metrics.

The Qualified observer complements the learned history-based estimate through an inertial error structure admitted by the calibrated, innovation-conditioned correction channel. Across the matched closed-loop cases, this arrangement reduced lateral-error RMS by 28.23%, heading-error RMS by 12.79%, and the mean input-increment index by 27.61%. The paired confidence intervals support the reductions in the two tracking measures, while all three matched means favor the fused configuration. Because the temporal estimate remains the primary source and every correction is bounded before scheduling, the fusion gain is obtained without transferring command generation to the learning or observer blocks. The resulting structure aligns with prior use of model-based information to manage grade-estimation uncertainty [14], [15], [39] and with hybrid physics–learning designs that retain the model-based controller as the command-generating mechanism [37].

#### VI. CONCLUSION

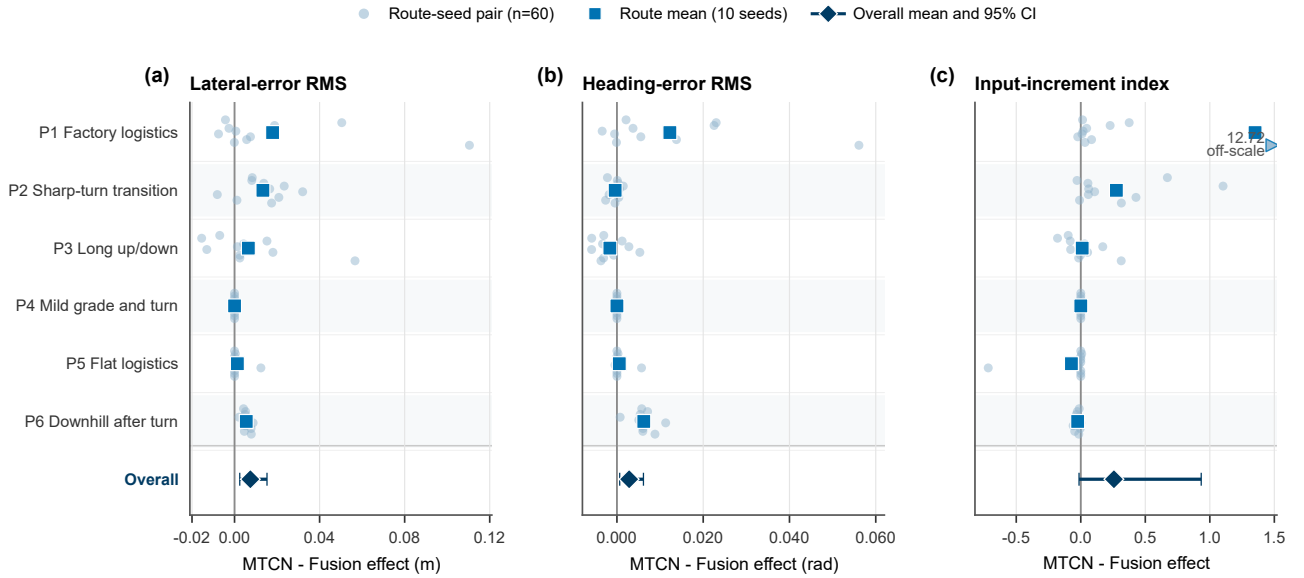
This study developed a slope-aware perception-to-control chain for a diagonal dual-steering-wheel AGV. The framework integrates history-based grade estimation, bounded assistance from the Qualified observer, and synchronized grade-aware LPV-MPC scheduling.

Compared with direct inertial-sensor-based grade estimation [13], [26], [27], ModernTCN-delta estimates road grade from history-only AGV responses without requiring a direct grade input. Its temporal backbone was adapted from the established ModernTCN architecture [23] through the dynamics-informed feature set and multi-scale temporal differences. In the ten-seed offline comparison, ModernTCN-delta achieved the lowest grade MAE and P95 absolute error of  $0.59888^\circ$  and  $1.67777^\circ$ , compared with  $0.65300^\circ$  and  $1.80626^\circ$  for ModernTCN-22D,  $0.88583^\circ$  and  $2.25950^\circ$  for GRU-22D [40], and  $0.91537^\circ$  and  $2.63580^\circ$  for TCN-22D [20]. After closed-loop deployment, ModernTCN-delta retained the lowest all-path mean MAE and P95 absolute error of  $0.423^\circ$  and  $1.237^\circ$ , respectively. Under the common CPU inference backend, its p95 core inference time was 0.4038 ms, while all four tested estimators remained below 1.2 ms.

Compared with MTCN-LPV-MPC using the AGV-adapted ModernTCN-delta estimator alone, Fusion-LPV-MPC adds a calibrated and bounded correction from the Qualified observer while retaining ModernTCN-delta as the primary grade source. Across the 60 matched route–seed cases, Fusion-

**TABLE 11. Matched MTCN-LPV-MPC and Fusion-LPV-MPC results over 60 route-seed pairs. Effects are defined as MTCN-LPV-MPC minus Fusion-LPV-MPC; boldface intervals exclude zero.**

| Metric             | MTCN-LPV-MPC | Fusion-LPV-MPC | Relative reduction | Paired effect [95% CI]            |
|--------------------|--------------|----------------|--------------------|-----------------------------------|
| $e_y$ RMS (m)      | 0.02635      | 0.01891        | 28.23%             | <b>0.00744 [0.00245, 0.01527]</b> |
| $e_\psi$ RMS (rad) | 0.02216      | 0.01933        | 12.79%             | <b>0.00283 [0.00064, 0.00615]</b> |
| $J_{\Delta u}$     | 0.93053      | 0.67361        | 27.61%             | 0.25692 [−0.01349, 0.93333]       |

**FIGURE 8. Matched closed-loop effects of the bounded fusion interface over 60 route-seed pairs. Panels show MTCN-LPV-MPC minus Fusion-LPV-MPC effects for (a) lateral-error RMS, (b) heading-error RMS, and (c) the composite input-increment index. Light markers denote individual route-seed pairs, blue squares denote route-level means over ten seeds, and dark diamonds with whiskers denote the overall paired means with 95% bootstrap confidence intervals. Positive effects favor Fusion-LPV-MPC. Exact ties remain at zero; the single distant  $J_u$  case is labelled off-scale, while its overall interval is shown on the main axis. Aggregate values are reported in Table 11.****TABLE 12. Same-backend core inference time and model size for the four tested estimators. p50/p95/p99 denote latency percentiles over 10000 timed calls.**

| Method          | Params (k) | Model (kB) | p50 (ms) | p95 (ms) | p99 (ms) | > 10 ms |
|-----------------|------------|------------|----------|----------|----------|---------|
| GRU-22D         | 99.276     | 400.682    | 0.3071   | 0.3377   | 0.4124   | 0/10000 |
| ModernTCN-22D   | 118.538    | 472.135    | 0.3582   | 0.3881   | 0.4942   | 0/10000 |
| ModernTCN-delta | 145.202    | 578.793    | 0.3744   | 0.4038   | 0.4818   | 0/10000 |
| TCN-22D         | 156.396    | 627.487    | 1.0452   | 1.1341   | 1.3597   | 1/10000 |

LPV-MPC reduced mean lateral-error RMS from 0.02635 m to 0.01891 m and mean heading-error RMS from 0.02216 rad to 0.01933 rad, corresponding to reductions of 28.23% and 12.79%, respectively. The paired effects were 0.00744 m (95% CI [0.00245, 0.01527]) and 0.00283 rad (95% CI [0.00064, 0.00615]). The mean input-increment index was also reduced by 27.61%, from 0.93053 to 0.67361. All 60 cases completed the prescribed execution checks, with exact fallback identity and runtime separation from ground-truth grade preserved.

Compared with LPV-MPC path-tracking formulations that omit grade scheduling [6], [28], [29], the proposed formulation synchronously applies one conditioned grade signal to the prediction model, measured-disturbance channel, grade feedforward, scheduled weights, and constraints. In the six-route experiment, zero-grade scheduling produced mean lateral-error RMS, heading-error RMS, and  $J_{\Delta u}$  val-

ues of 1.48046 m, 0.34204 rad, and 696.014, respectively. Causal scheduling with the Qualified observer reduced these values to 0.02051 m, 0.02347 rad, and 0.51646, while the analysis-only truth-driven reference further reduced them to 0.01248 m, 0.01139 rad, and 0.42684. All six route-paired confidence intervals for the Qualified-observer and truth-driven comparisons excluded zero. These results establish the control relevance of grade scheduling, show that a causal grade estimate recovers most of the closed-loop benefit, and indicate the additional performance potential provided by higher scheduling accuracy.

Future work will extend the present simulation evidence through physical-vehicle and hardware-in-the-loop validation, broader route, load, and measurement- disturbance conditions, and formal stability and error-bound analysis of the grade-scheduled LPV-MPC.

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